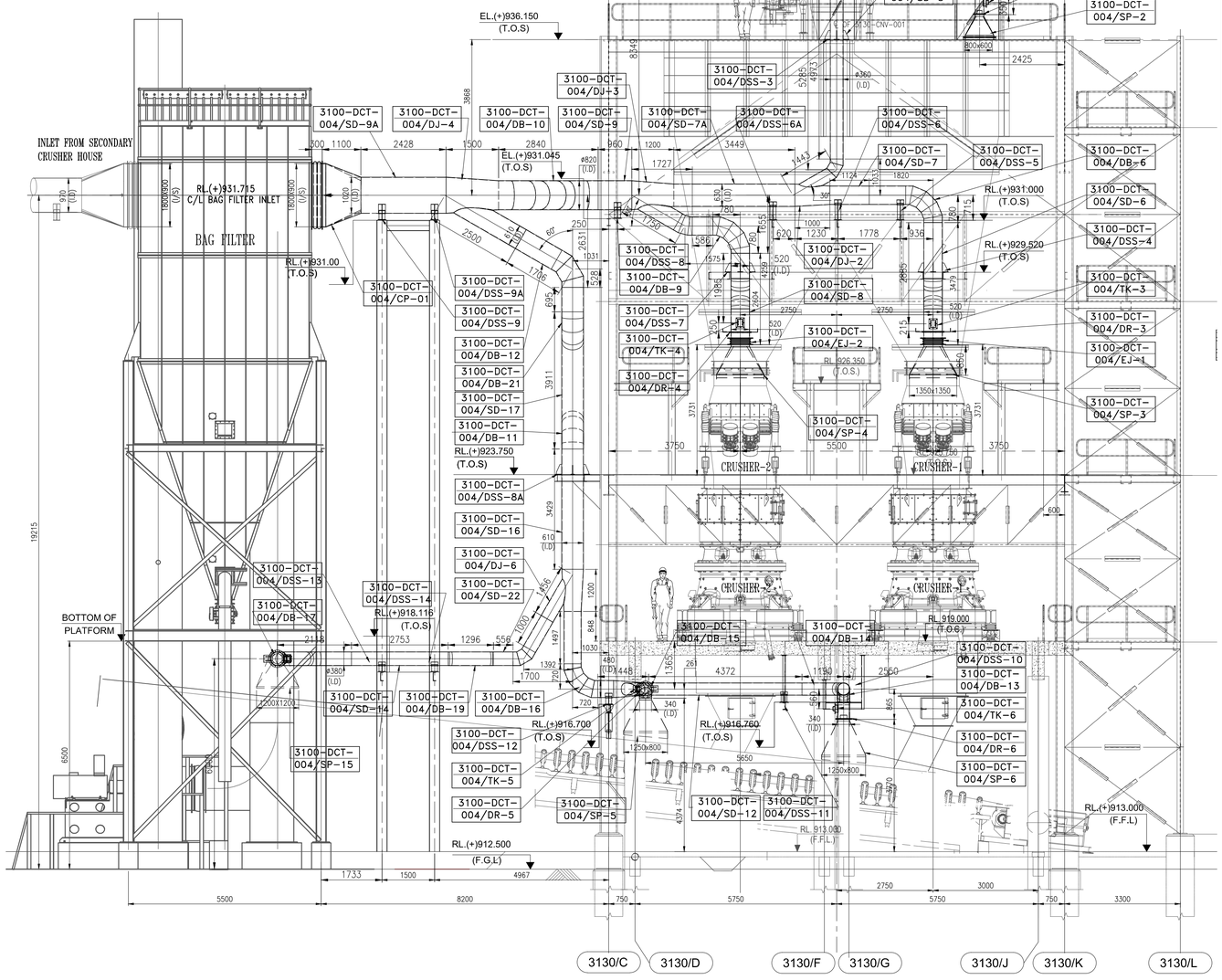
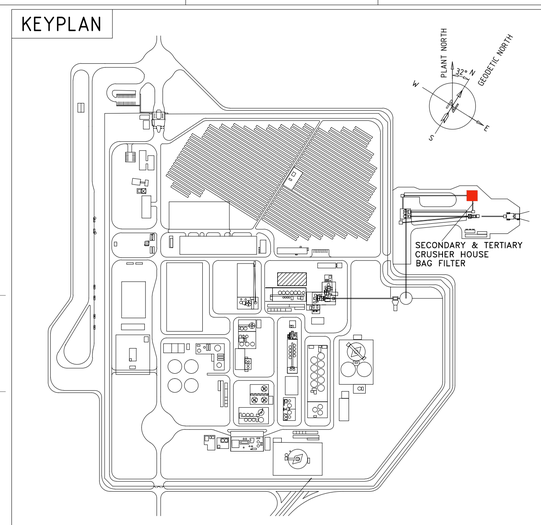
DRAWING NUMBER: MD-216-3100-EG-ME-GAD-1029 | REV: A02

PAINTING AND PROTECTIVE COATINGS	
COATING SYSTEM	2 – 2 AS PER MD-101-SMEM-EG-ME-SPC-0007_C01 (TABLE A-1 ITEM NO. 5: DUCT)
COLOUR CODE	6032 – Signal Green as per MD-101-SMEM-EG-ME-SPC-0007_C01 (Appendix B – Equipment Colour Coding: Dust collection system)



VIEW A-A
(FROM DRG.NO.-1030)

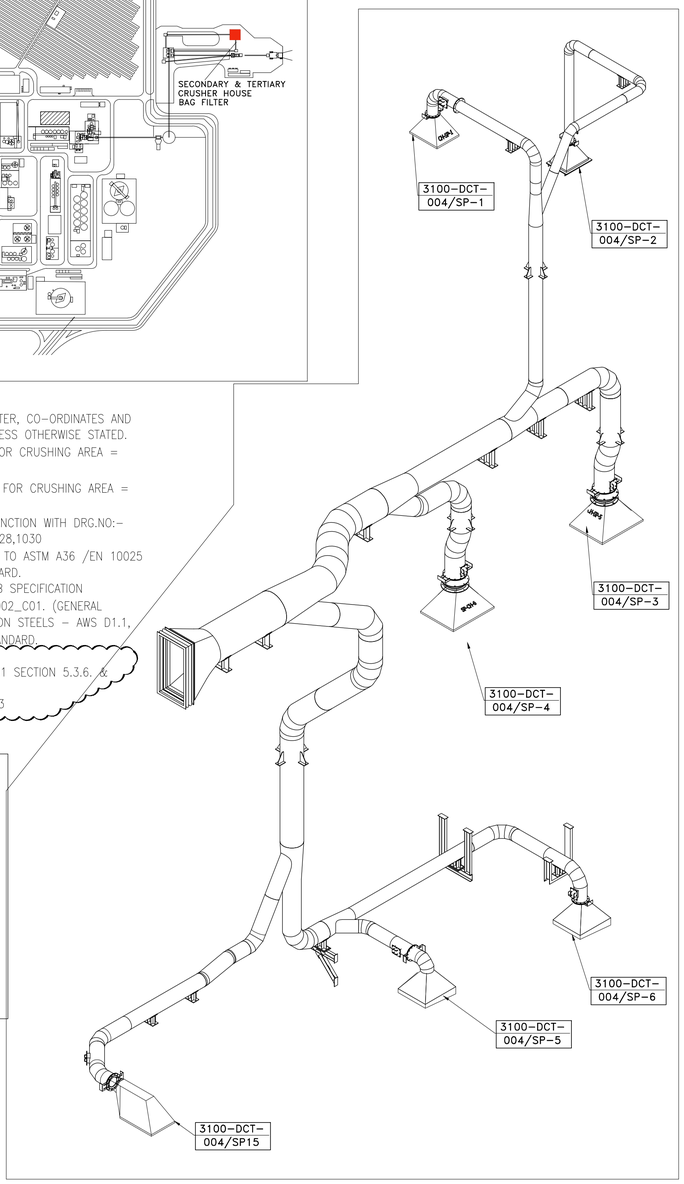
KEYPLAN



NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. THIS DRG.TO BE READ IN CONJUNCTION WITH DRG.NO:- MD-216-3100-EG-ME-GAD-1028,1030
5. DUCT MATERIAL SHALL CONFORM TO ASTM A36 /EN 10025 S275 JR OR EQUIVALENT STANDARD.
6. WELDING SHALL CONFORM TO ITB SPECIFICATION MD-101-SMEM-EG-GE-SPC-0002_C01. (GENERAL MECHANICAL FABRICATION: CARBON STEELS – AWS D1.1, AWS D1.3) OR EQUIVALENT STANDARD.
A02
7. DUCTS TO BE DESIGNED TO MD-101-SMEM-EG-ME-CRT-0001 SECTION 5.3.6. & APPROVED DESIGN CRITERIA MD-216-3100-EG-ME-CRT-1003

ABBREVIATION

SP-SUCTION POINT
SD-STRAIGHT DUCT
DB-DUCT BEND
DJ-DUCT JUNCTION
DR-MANUAL DAMPER(BAFFLE)
TK-INSPECTION COVER
DS-DUCT SUPPORT
DSS-DUCT SADDLE SUPPORT
SCH-SECONDARY CRUSHER HOUSE
CP-CHANGE PIECE
DE-DUST EXTRACTION
SYS-SYSTEM
TER-TERTIARY



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3100-EG-PR-PID-V509-2001	P&ID DUST EXTRACTION SYSTEM FOR SEC.& TER.CRUSHER HOUSE								
MD-216-3100-EG-ME-LAD-V509-2002	LAYOUT OF BAG FILTER & ID FAN								
MD-2016-3210-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHED ORE STORAGE								
MD-2016-3130-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING TER. CRUSHING								
MD-2016-3120-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING SEC. CRUSHING								
MD-2016-3110-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM CRUSHING PRIMARY CRUSHING	A02	17/07/2020	REVISED AS PER CUSTOMER COMMENT DATED-15/06/2020	TKS	SYR	ABD	SBA/JPD	
MD-316-3100-EG-ME-SDD-1001	SCHEMATIC DIAGRAM CRUSHING DUST EXTRACTION	A01	12/05/2020	ISSUED FOR APPROVAL	TKS	ABD	ABD	SBA/JPD	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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Saudi Arabian Mining Company شركة معادن العربية السعودية

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LARSEN & TOUBRO DRAWING NUMBER	O19045_080_U_G_21417_002	REVISION 1
OUTOTEC DRAWING NUMBER	900707 _GAU01 _ZC01 _MTL01 _31017	DOCUMENT ID OU100672978 / REVISION 1

DRAWN	TKS	08/05/2020
CHECKED	ABD	10/05/2020
ENGINEER	ABD	10/05/2020
APPROVED	SBA/JPD	12/05/2020
EMPLOYER		
SCALE	1:75	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
CRUSHING
DUCT-DUST EXTRACTION SYSTEM FOR TERTIARY CRUSHER HOUSE

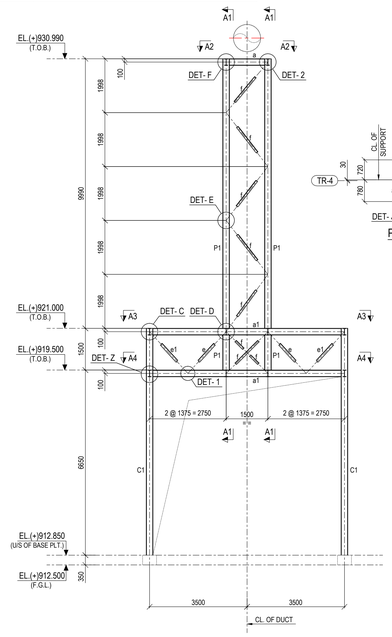
DRAWING NUMBER: MD-216-3100-EG-ME-GAD-1029 | REV: A02

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

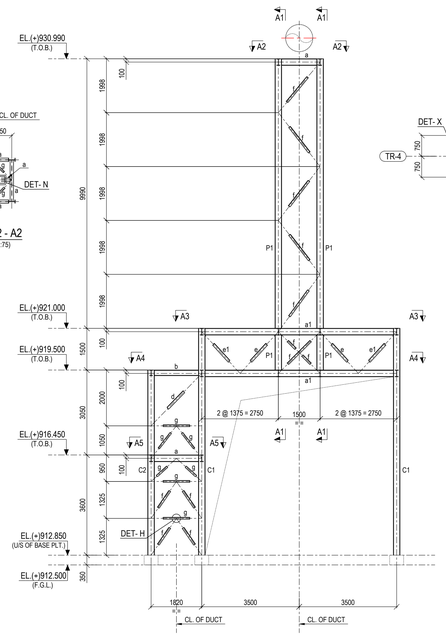
Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	18-08-2020	21-10-2020	12-10-2023				Date	18-08-2020	21-10-2020	12-10-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 2 **STRUCTURAL GENERAL ARRANGEMENT DRAWINGS CONTAINING BASE PLATE DETAIL**

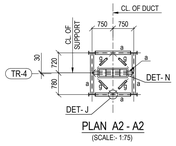
DRAWING NUMBER: MD-216-3120-EG-ST-GAD-2021 REV A01



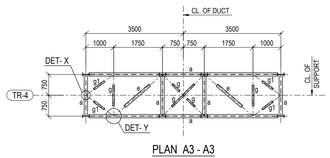
VIEW A - A
(SCALE:- 1:75)



VIEW B - B
(SCALE:- 1:75)



PLAN A2 - A2
(SCALE:- 1:75)



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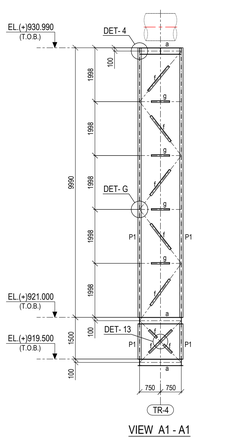
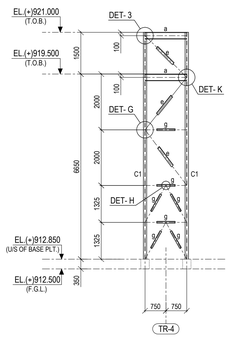
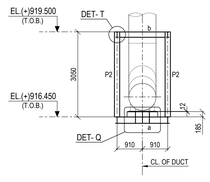


VIEW A1 - A1
(SCALE:- 1:75)



VIEW C - C
(SCALE:- 1:75)



VIEW A6 - A6
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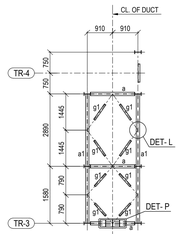

PLAN A5 - A5
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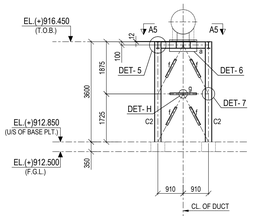

VIEW F - F
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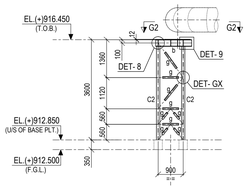
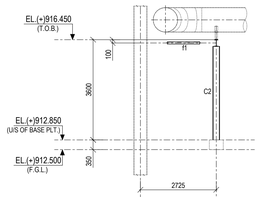


VIEW G - G
(SCALE:- 1:75)



VIEW G1 - G1
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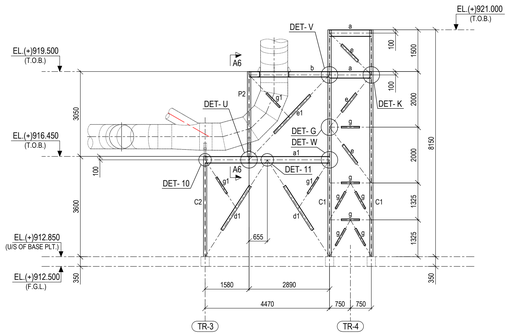
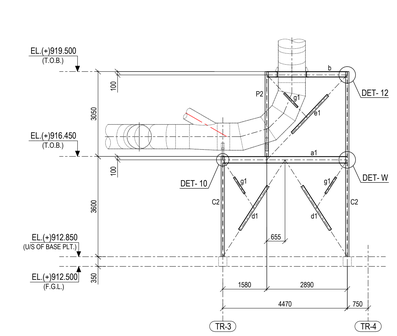


VIEW D - D
(SCALE:- 1:75)



VIEW E - E
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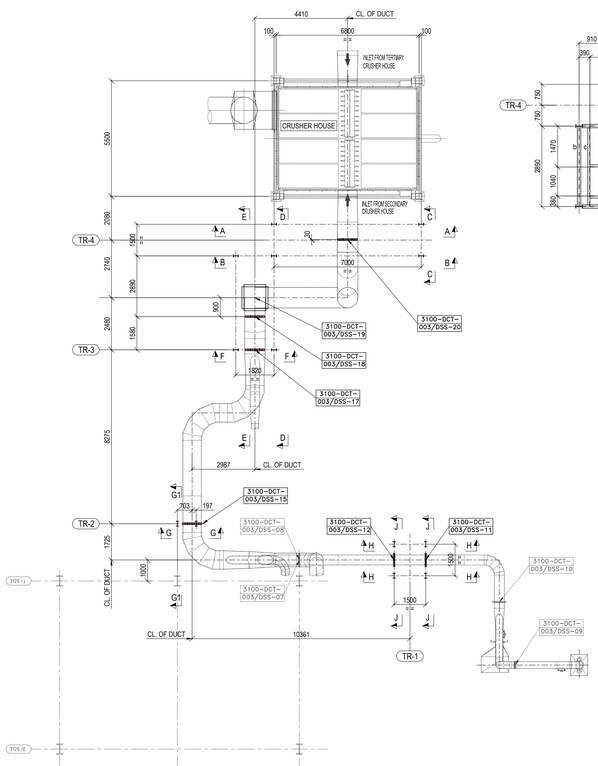

LAYOUT OF SECONDARY DUCT SUPPORT
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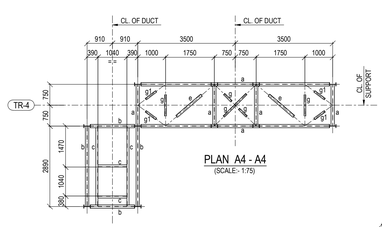
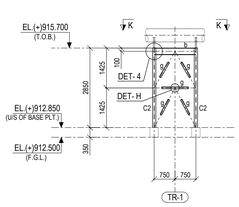
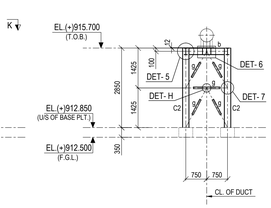


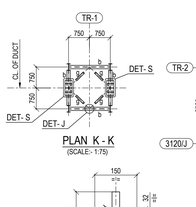
PLAN A4 - A4
(SCALE:- 1:75)



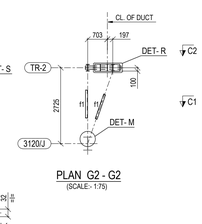
VIEW H - H
(SCALE:- 1:75)



VIEW J - J
(SCALE:- 1:75)



PLAN K - K
(SCALE:- 1:75)



PLAN G2 - G2
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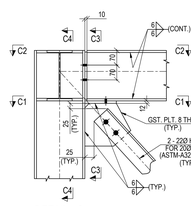

DETAIL - C
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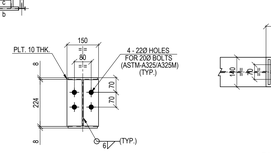

VIEW C3 - C3
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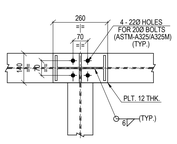
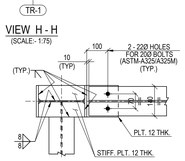


PLAN D1 - D1
(SCALE:- 1:10)



PLAN C1 - C1
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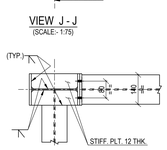

PLAN C2 - C2
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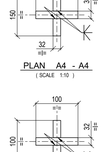
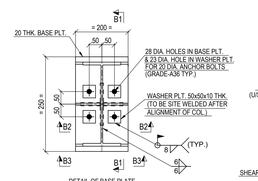
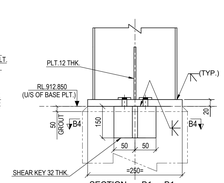
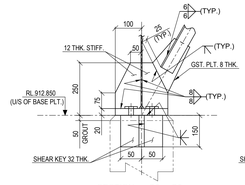




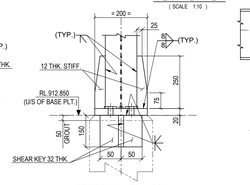
DETAIL OF BASE PLATE
(FOR IPE 200 COLUMN)
(SCALE 1:10)



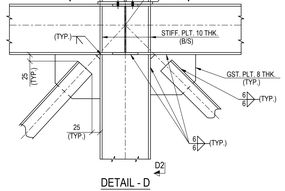
SECTION B1 - B1
(SCALE 1:10)



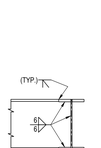
SECTION B2 - B2
(SCALE 1:10)

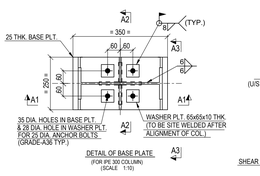


SECTION B3 - B3
(SCALE 1:10)

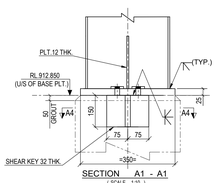


DETAIL - D
(SCALE:- 1:10)

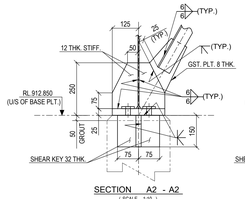




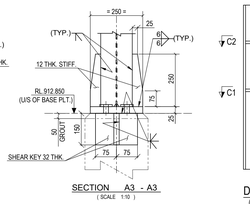
DETAIL OF BASE PLATE
(FOR IPE 300 COLUMN)
(SCALE 1:10)



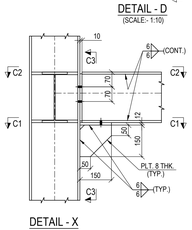
SECTION A1 - A1
(SCALE 1:10)



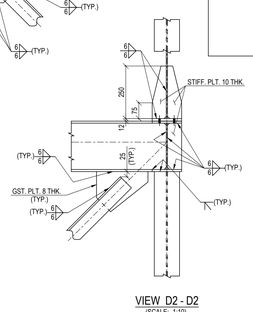
SECTION A2 - A2
(SCALE 1:10)



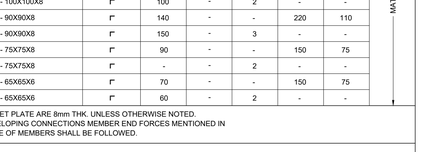
SECTION A3 - A3
(SCALE 1:10)



DETAIL - X
(SCALE:- 1:10)



VIEW D2 - D2
(SCALE:- 1:10)



TYP DETAIL OF WELDED CONN.
(SCALE:- 1:10)

TYP DETAIL OF BOLTTED CONN.
(SCALE:- 1:10)

SCHEDULE OF MEMBERS

MEMBER MKD.	MEMBER	SECTON	FACTORED AXIAL FORCE(KN)	FACTORED SHEAR FORCE(KN)	NO. OF BOLTS.	WELD LENGTH (mm) (HEEL SIDE) L1	WELD LENGTH (mm) (TOE SIDE) L2	REMARKS
C1	IPE 300	I	-	-	-	-	-	MATERIAL GRADE:S275 JR
C2	IPE 200	I	-	-	-	-	-	
P1	IPE 240	I	-	-	-	-	-	
P2	IPE 200	I	-	-	-	-	-	
a	IPE 240	I	150	70	-	-	-	
a1	1/WEB PLT. 220x8 THK. 2/FLG. PLT. 140x10 THK.	I	150	70	-	-	-	
b	IPE 200	I	20	50	-	-	-	
c	UPN 200	[	-	-	-	-	-	
d	L - 100X100X8	r	100	-	-	175	100	
d1	L - 100X100X8	r	100	-	2	-	-	
e	L - 90X90X8	r	140	-	-	220	110	
e1	L - 90X90X8	r	150	-	3	-	-	
f	L - 75X75X8	r	90	-	-	150	75	
f1	L - 75X75X8	r	-	-	2	-	-	
g	L - 65X65X6	r	70	-	-	150	75	
g1	L - 65X65X6	r	60	-	2	-	-	

ALL GUSSET PLATE ARE 8mm THK. UNLESS OTHERWISE NOTED.
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.

NOTES:

1. ALL DIMENSIONS & CO-ORDINATES ARE IN MILLIMETER AND LEVELS ARE IN METER UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36.
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
4. ALL WELDS SHALL BE OF SHOP QUALITY & TO BE TESTED AS PER QUALITY ASSURANCE PLAN.
5. ALL PERMANENT STRUCTURAL BOLTS CONFORM TO ASTM A 325M.
6. PAINTING ON STRUCTURES WILL BE AS PER DESIGN BASIS REPORT (DBR).

LEGENDS:-

TOS - TOP OF BEAM
U/S - UNDER SIDE
CL - CENTER LINE
TYP. - TYPICAL
PLT. - PLATE
THK. - THICKNESS
FGL - FINISHED GROUND LEVEL
B/S - BOTH SIDE
GST. - GUSSET
STIFF. - STIFFNER
CONT. - CONTINUE
CONN. - CONNECTION

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-ST-CRT-1001	CIVIL,STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA	A01	10/08/2020	ISSUED FOR REFERENCE	HSK	SBY	RPK	PRI	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: O19045_080_S_G_27013_002 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31286 — DOCUMENT ID: OU100692120 — REVISION: 0

DRAWN	HSK	01/08/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SBY	03/08/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	RPK	04/08/2020	SECONDARY CRUSHING
APPROVED	PRI	06/08/2020	PLAN, ELEVATION AND DETAIL OF DEDUSTING DUCT SUPPORT
EMPLOYER			

SCALE: 1 : 10, 75 & 100 — A0 — DRAWING NUMBER: MD-216-3120-EG-ST-GAD-2021 — REV: A01

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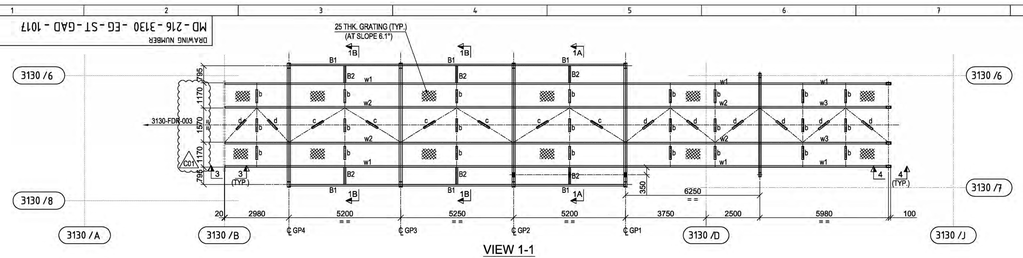
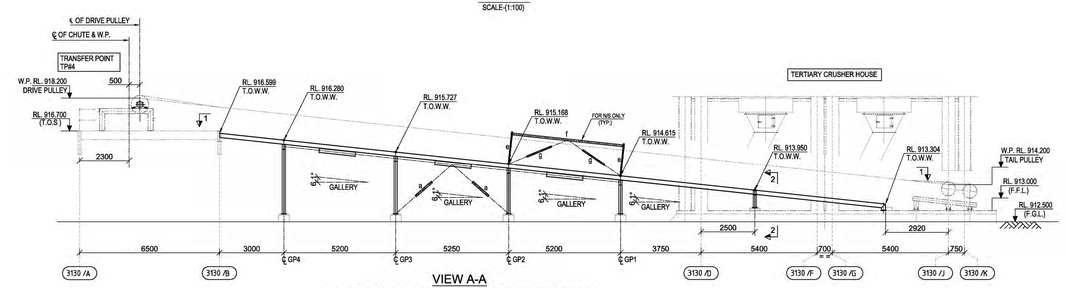
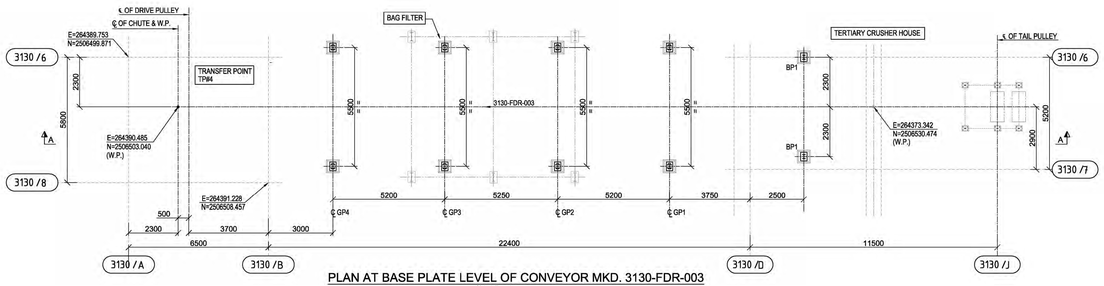
DRAWING NUMBER: MD-216-3130-EG-ST-GAD-1017 | REV: C01



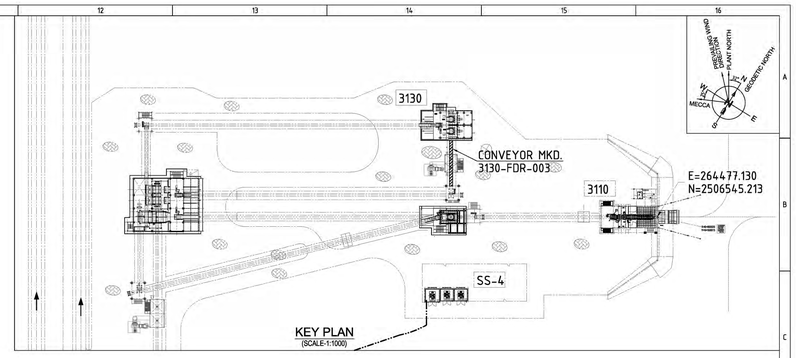
VIEW 1-1
SCALE-(1:100)

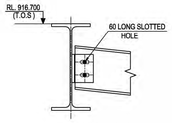


VIEW A-A
ELEVATION OF CONVEYOR MKD. 3130-FDR-003
(SCALE 1:100)

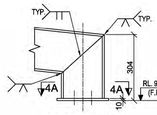


PLAN AT BASE PLATE LEVEL OF CONVEYOR MKD. 3130-FDR-003
(SCALE 1:100)

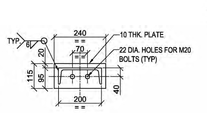




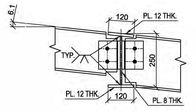
SECTION 3-3
(SCALE 1:10)



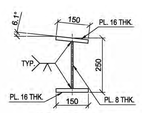
SECTION 4-4
(SCALE 1:10)



SECTION 4A-4A
(SCALE 1:10)



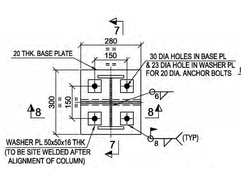
SECTION 5-5
(SCALE 1:10)



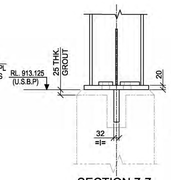
SECTION 6-6
(SCALE 1:10)

SCHEDULE OF MEMBERS					
MEMBER MARKED	MEMBER	SECTON	FACTORED AXIAL FORCE(KN)	FACTORED SHEAR FORCE(KN)	REMARKS
C1	IPE 240	I	-	-	MATERIAL GRADE:E275-JR
C2	1. PL. 150X16 2. PL. 218X8	1 2 1	-	-	
C3	1. PL. 120X12 2. PL. 226X8	1 2 1	-	-	
B1	UPN 200	[	-	-	
B2	IPE 200	I	-	-	
a	2L 75X75X8	T	95	-	
b	L 50X50X6	L	10	-	
c	L 75X75X8	L	20	-	
d	L 65X65X6	L	20	-	
e	UPN 100	[	-	-	
f	L 75X75X8	L	-	-	
g	L 50X50X6	L	-	-	
w3	UPN 240	[	-	-	
w2	UPN 200	[	-	-	
w1	UPN 140	[	-	-	

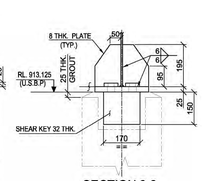
ALL GUSSET PLATE ARE 8mm THK.
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED FOR WELDED CONNECTION. FOR BOLTED CONNECTION, MINIMUM 2 Nos. M20 BOLTS SHALL BE PROVIDED.



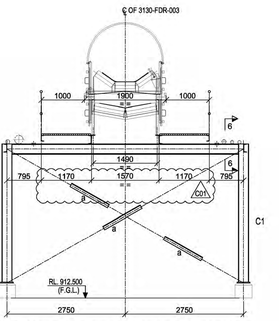
BASE PLATE DETAIL FOR BP1
(SCALE - 1:10)



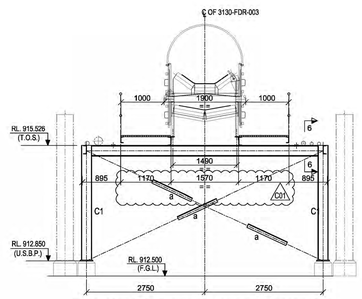
SECTION 7-7
(SCALE - 1:10)



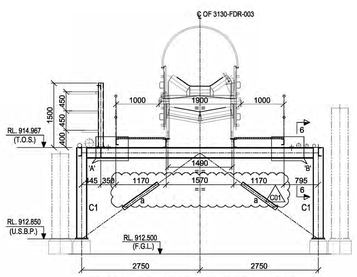
SECTION 8-8
(SCALE - 1:10)



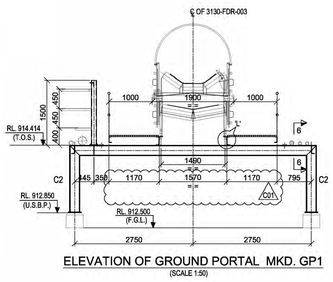
ELEVATION OF GROUND PORTAL MKD. GP4
(SCALE 1:50)



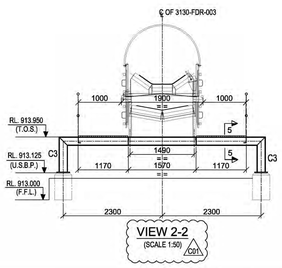
ELEVATION OF GROUND PORTAL MKD. GP3
(SCALE 1:50)



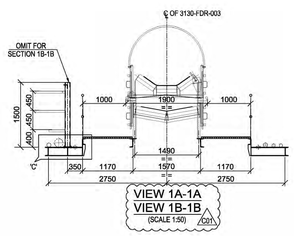
ELEVATION OF GROUND PORTAL MKD. GP2
(SCALE 1:50)



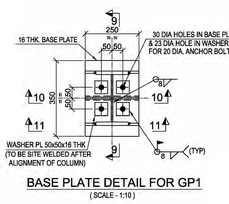
ELEVATION OF GROUND PORTAL MKD. GP1
(SCALE 1:50)



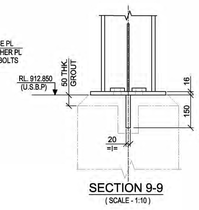
VIEW 2-2
(SCALE 1:50)



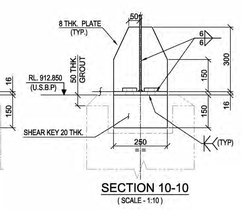
VIEW 1A-1A
VIEW 1B-1B
(SCALE 1:50)



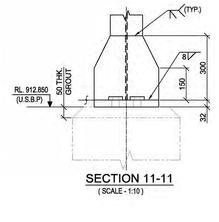
BASE PLATE DETAIL FOR GP1
(SCALE - 1:10)



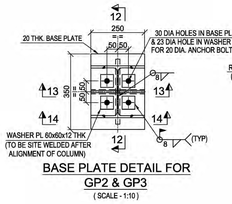
SECTION 9-9
(SCALE - 1:10)



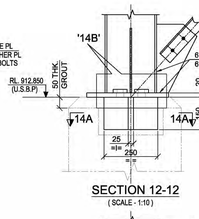
SECTION 10-10
(SCALE - 1:10)



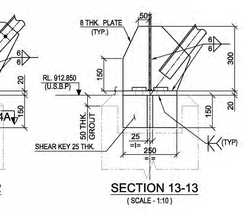
SECTION 11-11
(SCALE - 1:10)



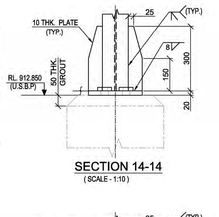
BASE PLATE DETAIL FOR GP2 & GP3
(SCALE - 1:10)



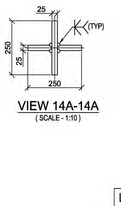
SECTION 12-12
(SCALE - 1:10)



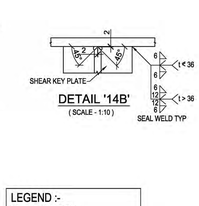
SECTION 13-13
(SCALE - 1:10)



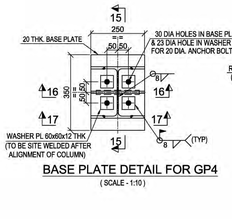
SECTION 14-14
(SCALE - 1:10)



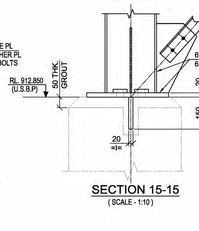
VIEW 14A-14A
(SCALE - 1:10)



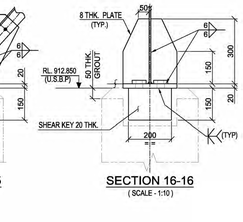
DETAIL '14B'
(SCALE - 1:10)



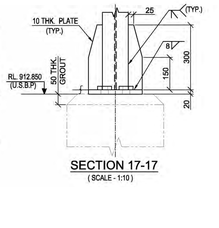
BASE PLATE DETAIL FOR GP4
(SCALE - 1:10)



SECTION 15-15
(SCALE - 1:10)



SECTION 16-16
(SCALE - 1:10)



SECTION 17-17
(SCALE - 1:10)

LEGEND :-
R.L - REDUCED LEVEL
℄ - CENTRE LINE
T.O.S. - TOP OF STEEL
F.G.L. - FINISHED GROUND LEVEL.
F.F.L. - FINISHED FLOOR LEVEL.
TYP. - TYPICAL.
MKD. - MARKED.
THK. - THICK.
F/S - FAR SIDE
N/S - NEAR SIDE
W.P.-WORKING POINT.
U.S.B.P. - UNDER SIDE OF BASE PLATE.
U.N.O. - UNLESS NOTED OTHERWISE
T.O.W.W. - TOP OF WALKWAY
REF - REFER
DRG. NO. - DRAWING NUMBER

NOTES :-
1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS & COORDINATES ARE IN METER UNO.
2. GRADE OF STEEL USED IS S275R AS PER EN10025.
3. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD - 216 - 3130 -EG-ST - GAD - 1018 & MD - 216 - 3130 -EG -ST - GAD - 1019.
4. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO MD - 216 - 1000 -EG -ST - CRT - 1001

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-ELV-1015	FEEDER 3130-FDR-004 - ELEVATION								
MD-216-3130-EG-ME-LPX-1008	FEEDER 3130-FDR-004 - PLAN	C01	04/05/2020	ISSUED FOR CONSTRUCTION, REVISED AS PER EMPLOYER					
MD-216-3130-EG-ME-ELV-1013	FEEDER 3130-FDR-003 -ELEVATION			COMMENTS DATED 27/04/2020 & REVISED AS MARKED	SUS	SBY	APR	PRI	
MD-216-3130-EG-ME-LPX-1007	FEEDER 3130-FDR-003 - PLAN	A01	03/04/2020	ISSUED FOR REFERENCE	SUS	SBY	APR	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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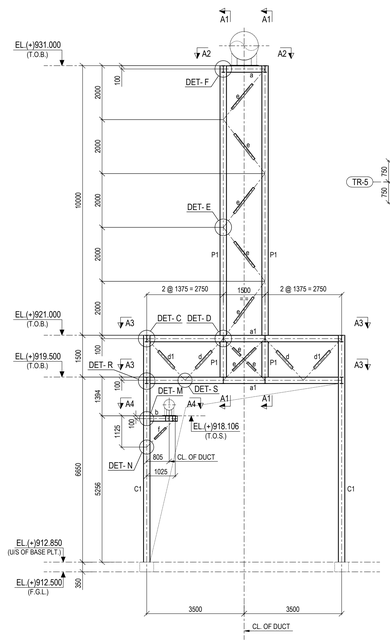
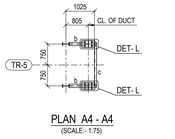
LARSEN & TOUBRO DRAWING NUMBER	019045_100_S_G_26013_029		REVISION 1
OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31308	DOCUMENT ID OU100668060	REVISION 1
DRAWN	SUS	01/04/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SBY	02/04/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	APR	01/04/2020	STRUCTURAL GA FOR 3130-FDR-003 & 004, SH1
APPROVED	PRI	03/04/2020	PLAN, ELEVATIONS & DETAILS
EMPLOYER			
SCALE	1:100,50,20,10	A0	DRAWING NUMBER: MD - 216 - 3130 - EG - ST - GAD - 1017 | REV: C01

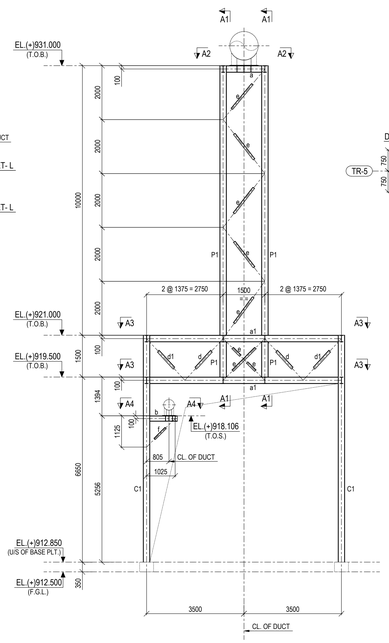
DRAWING NUMBER: MD-216-3130-EG-ST-GAD-2020 | REV: A01



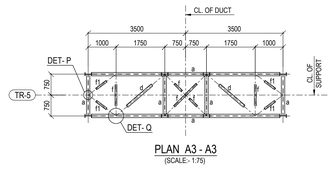
VIEW A - A
(SCALE:- 1:75)



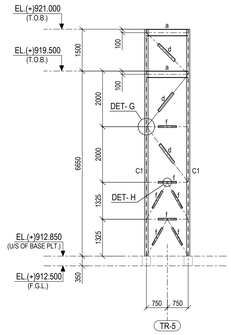
PLAN A4 - A4
(SCALE:- 1:75)



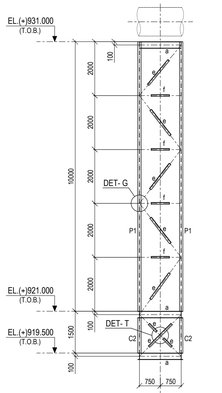
VIEW B - B
(SCALE:- 1:75)



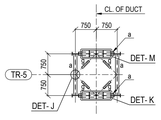
PLAN A3 - A3
(SCALE:- 1:75)



VIEW C - C
(SCALE:- 1:75)



VIEW A1 - A1
(SCALE:- 1:75)



PLAN A2 - A2
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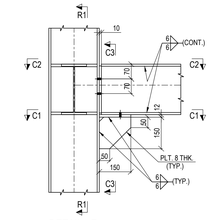
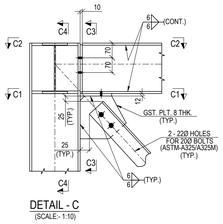
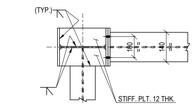


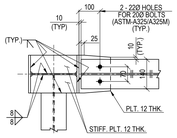
DETAIL - R
(SCALE:- 1:10)



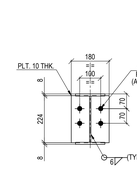
DETAIL - C
(SCALE:- 1:10)



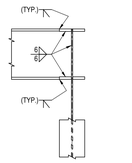
PLAN C2 - C2
(SCALE:- 1:10)



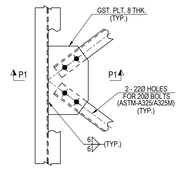
PLAN C1 - C1
(SCALE:- 1:10)



VIEW C3 - C3
(SCALE:- 1:10)

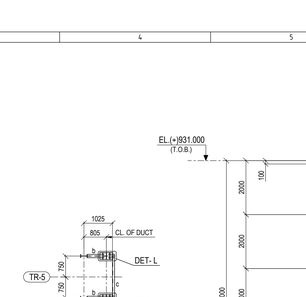


VIEW C4 - C4
(SCALE:- 1:10)

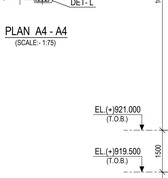


DETAIL - P
(SCALE:- 1:10)

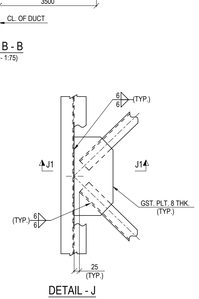
LAYOUT OF TERTIARY DUCT SUPPORT
(SCALE:- 1:100)



DETAIL - D
(SCALE:- 1:10)



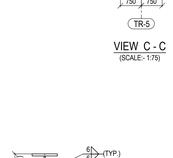
PLAN D1 - D1
(SCALE:- 1:10)



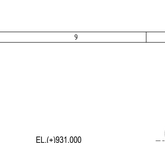
DETAIL - J
(SCALE:- 1:10)

VIEW D2 - D2
(SCALE:- 1:10)

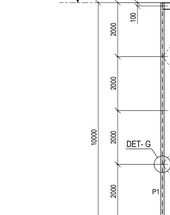
DETAIL - E
(SCALE:- 1:10)



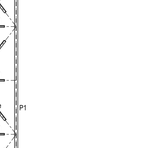
VIEW J1 - J1
(SCALE:- 1:10)



DETAIL - T
(SCALE:- 1:10)

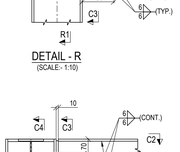


DETAIL - G
(SCALE:- 1:10)

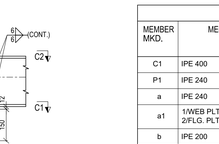


DETAIL - H
(SCALE:- 1:10)

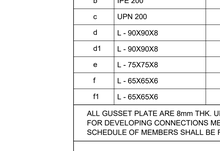
DETAIL - Q
(SCALE:- 1:10)



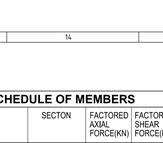
DETAIL - N
(SCALE:- 1:10)



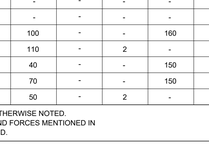
DETAIL - M
(SCALE:- 1:10)



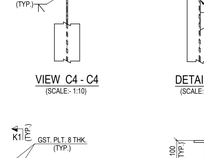
DETAIL - S
(SCALE:- 1:10)



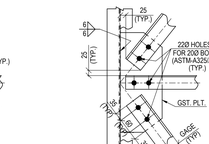
DETAIL - K
(SCALE:- 1:20)



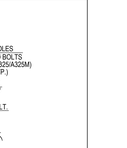
DETAIL - M
(SCALE:- 1:20)



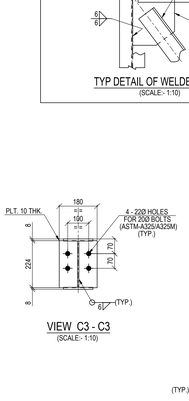
VIEW P1 - P1
(SCALE:- 1:10)



VIEW K1 - K1
(SCALE:- 1:10)



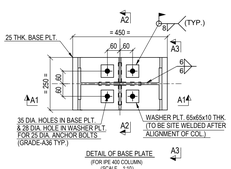
DETAIL - L
(SCALE:- 1:20)



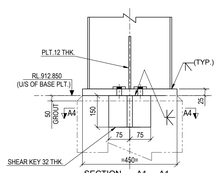
DETAIL - F
(SCALE:- 1:10)

VIEW F1 - F1
(SCALE:- 1:10)

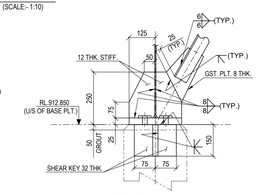
VIEW R1 - R1
(SCALE:- 1:10)



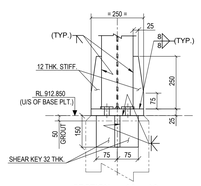
DETAIL OF BASE PLATE
(FOR PR 450 COLUMN)
(SCALE 1:10)



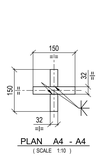
SECTION A1 - A1
(SCALE 1:10)



SECTION A2 - A2
(SCALE 1:10)



SECTION A3 - A3
(SCALE 1:10)

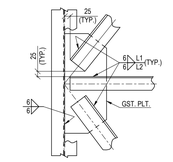


PLAN A4 - A4
(SCALE 1:10)

SCHEDULE OF MEMBERS

MEMBER MKD.	MEMBER	SECTON	FACTORED AXIAL FORCE(KN)	FACTORED SHEAR FORCE(KN)	NO. OF BOLTS,	WELD LENGTH (mm) (HEEL SIDE) L1	WELD LENGTH (mm) (TOE SIDE) L2	REMARKS
C1	IPE 400	I	-	-	-	-	-	MATERIAL GRADE:S275 JR
P1	IPE 240	I	-	-	-	-	-	
a	IPE 240	I	110	70	-	-	-	
a1	1/WEB PLT. 220x8 THK. 2/FLG. PLT. 140x10 THK.	I 6t 6t	110	70	-	-	-	
b	IPE 200	I	-	-	-	-	-	
c	UPN 200	C	-	-	-	-	-	
d	L - 90X90X8	L	100	-	-	160	90	
d1	L - 90X90X8	L	110	-	2	-	-	
e	L - 75X75X8	L	40	-	-	150	75	
f	L - 65X65X6	L	70	-	-	150	75	
f1	L - 65X65X6	L	50	-	2	-	-	

ALL GUSSET PLATE ARE 8mm THK. UNLESS OTHERWISE NOTED.
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.



TYP DETAIL OF WELDED CONN.
(SCALE:- 1:10)

TYP DETAIL OF BOLTTED CONN.
(SCALE:- 1:10)

NOTES:

1. ALL DIMENSIONS & CO-ORDINATES ARE IN MILLIMETER AND LEVELS ARE IN METER UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36.
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
4. ALL WELDS SHALL BE OF SHOP QUALITY & TO BE TESTED AS PER QUALITY ASSURANCE PLAN.
5. ALL PERMANENT STRUCTURAL BOLTS CONFORM TO ASTM A 325M.
6. PAINTING ON STRUCTURES WILL BE AS PER DESIGN BASIS REPORT (DBR).

LEGENDS:-

TOB - TOP OF BEAM
U/S - UNDER SIDE
CL - CENTER LINE
TYP. - TYPICAL
PLT - PLATE
THK. - THICKNESS
FGL - FINISHED GROUND LEVEL
B/S - BOTH SIDE
GST. - GUSSET
STIFF. - STIFFNER
CONT. - CONTINUE
CONN. - CONNECTION

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-ST-CRT-1001	CIVIL,STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA	A01	10/08/2020	ISSUED FOR REFERENCE	HSK	SBY	RPK	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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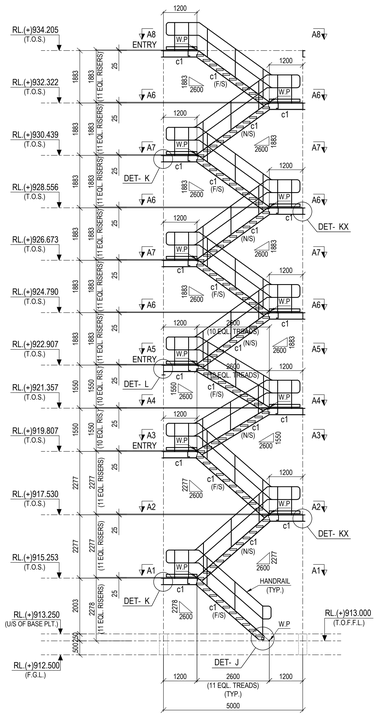
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31390 | DOCUMENT ID: QU100692074 | REVISION: 0

DRAWN	HSK	01/08/2020
CHECKED	SBY	03/08/2020
ENGINEER	RPK	04/08/2020
APPROVED	PRI	06/08/2020
EMPLOYER		

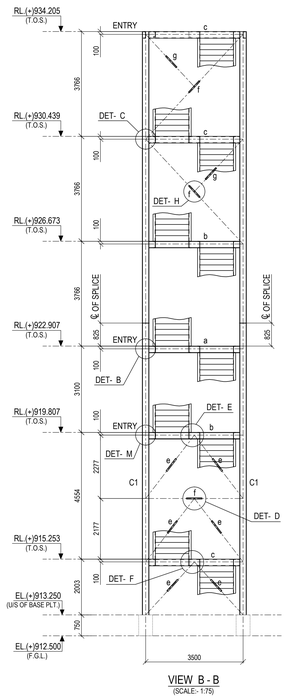
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
PLAN, ELEVATION AND DETAIL OF DEDUSTING DUCT SUPPORT

SCALE: 1 : 10, 20, 75 & 100 | A0
DRAWING NUMBER: MD - 216 - 3130 - EG - ST - GAD - 2020 | REV: A01

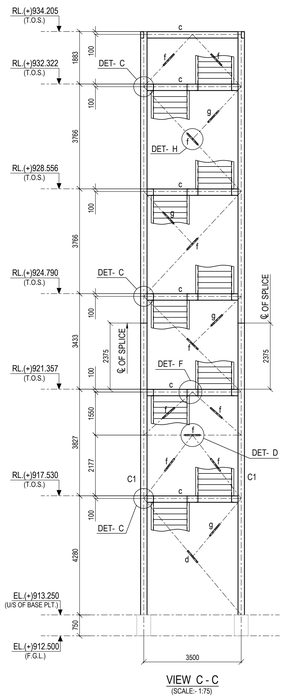
DRAWING NUMBER: MD-216-3380-EG-ST-GAD-1001 REV A01



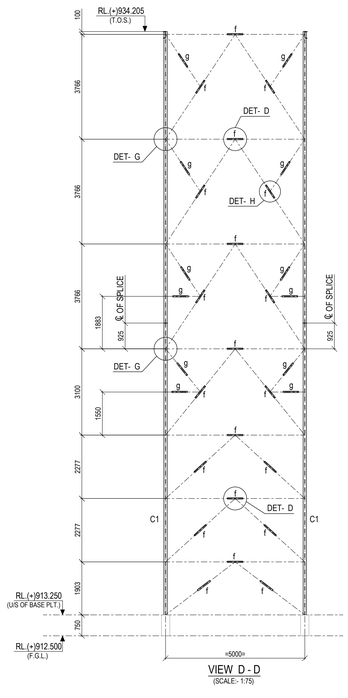
VIEW A - A
(SCALE:- 1:75)



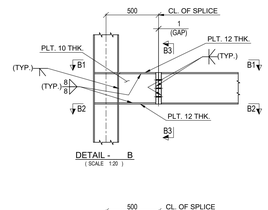
VIEW B - B
(SCALE:- 1:75)



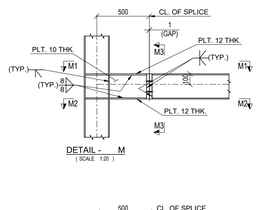
VIEW C - C
(SCALE:- 1:75)



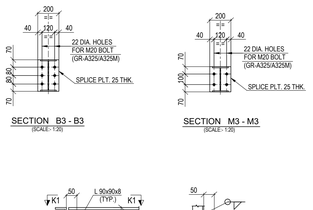
VIEW D - D
(SCALE:- 1:75)



DETAIL - B
(SCALE 1:20)

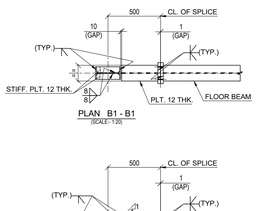


DETAIL - M
(SCALE 1:20)

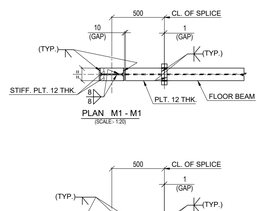


SECTION B3 - B3
(SCALE:- 1:20)

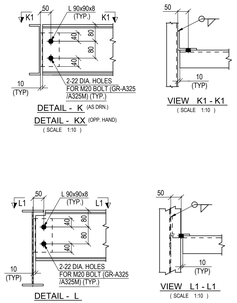
SECTION M3 - M3
(SCALE:- 1:20)



PLAN B1 - B1
(SCALE:- 1:20)



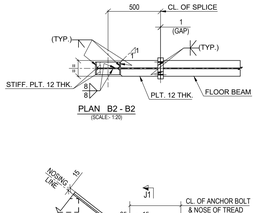
PLAN M1 - M1
(SCALE:- 1:20)



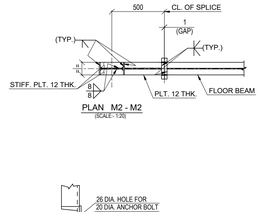
DETAIL - K (AS DRN.)

DETAIL - KX (OPP. HAND)
(SCALE 1:10)

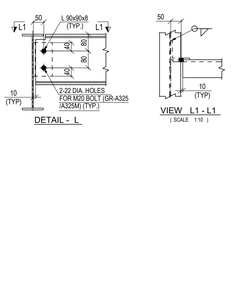
VIEW K1 - K1
(SCALE 1:10)



PLAN B2 - B2
(SCALE:- 1:20)

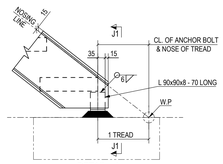


PLAN M2 - M2
(SCALE:- 1:20)

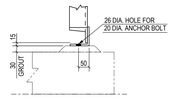


DETAIL - L

VIEW L1 - L1
(SCALE 1:10)



DETAIL - J
(SCALE 1:10)

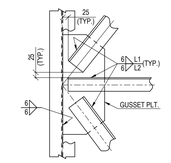


VIEW J1 - J1
(SCALE 1:10)

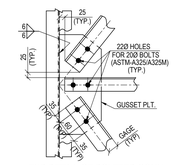
SCHEDULE OF MEMBERS

MEMBER MKD.	MEMBER	SECTON	FACTORED AXIAL FORCE(KN)	FACTORED SHEAR FORCE(KN)	NO. OF BOLTS.	WELD LENGTH (mm) (HEEL SIDE) L1	WELD LENGTH (mm) (TOE SIDE) L2	REMARKS
C1	IPE 240	I	-	-	-	-	-	MATERIAL GRADE:S275 JR
a	IPE300	I	10.0	70.0	6	-	-	
b	IPE240	I	70.0	50.0	4	-	-	
c	UPN240	[	70.0	50.0	4	-	-	
c1	UPN240	[	20.0	50.0	4	-	-	
d	L - 90X90X8	⊢	150.0	-	3	-	-	
e	L - 75X75X8	⊢	110.0	-	2	-	-	
f	L - 65X65X6	⊢	80.0	-	2	-	-	
g	L - 65X65X6	⊢	-	-	2	-	-	
h	UPN200	[	-	-	4	-	-	
j	L - 50X50X6	⊢	-	-	2	-	-	

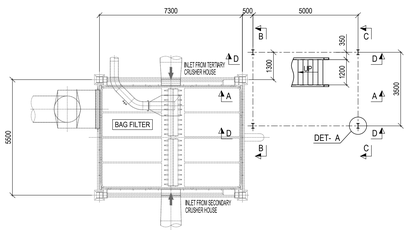
ALL GUSSET PLATE ARE 8mm THK. UNLESS OTHERWISE NOTED.
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.



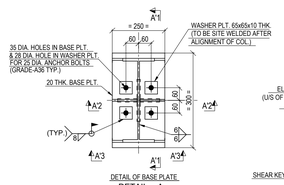
TYP DETAIL OF WELDED CONN.
(SCALE:- 1:10)



TYP DETAIL OF BOLTED CONN.
(SCALE:- 1:10)

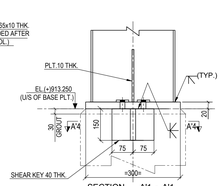


LAYOUT OF STAIR AT CRUSHER HOUSE BAG FILTER
(SCALE:- 1:100)

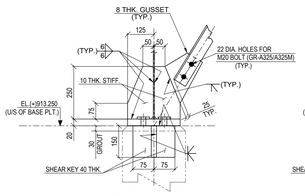


DETAIL - A
(SCALE 1:10)

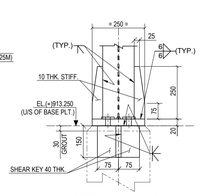
DETAIL OF BASE PLATE



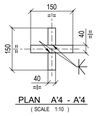
SECTION A'1 - A'1
(SCALE 1:10)



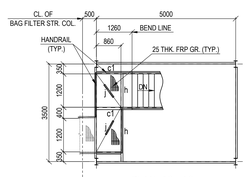
SECTION A'2 - A'2
(SCALE 1:10)



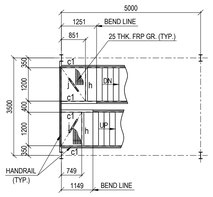
SECTION A'3 - A'3
(SCALE 1:10)



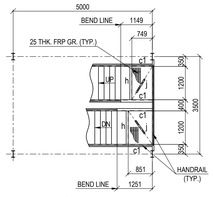
PLAN A'4 - A'4
(SCALE 1:10)



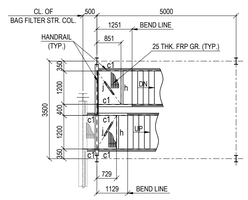
PLAN A8 - A8
(ALL PLAN BRACING ARE BELOW 100 MM. T.O.S)
(SCALE:- 1:75)



PLAN A1 - A1
(ALL PLAN BRACING ARE BELOW 100 MM. T.O.S)
(SCALE:- 1:75)



PLAN A2 - A2
(ALL PLAN BRACING ARE BELOW 100 MM. T.O.S)
(SCALE:- 1:75)



PLAN A3 - A3
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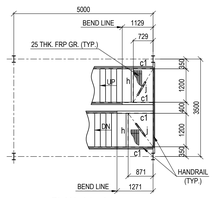
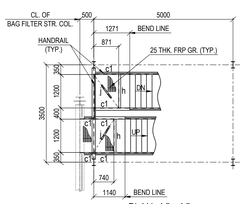
(SCALE:- 1:75)



PLAN A4 - A4
(ALL PLAN BRACING ARE BELOW 100 MM. T.O.S)
(SCALE:- 1:75)



PLAN A5 - A5
(ALL PLAN BRACING ARE BELOW 100 MM. T.O.S)
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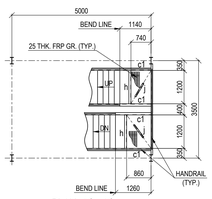

PLAN A6 - A6
(ALL PLAN BRACING ARE BELOW 100 MM. T.O.S)
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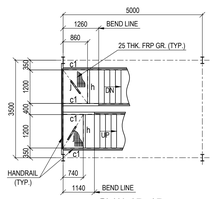

PLAN A7 - A7
(ALL PLAN BRACING ARE BELOW 100 MM. T.O.S)
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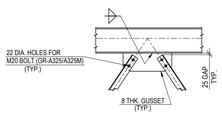
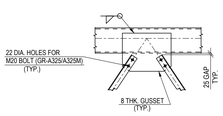
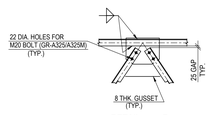


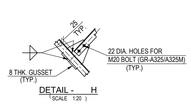
DETAIL - E
(SCALE 1:20)



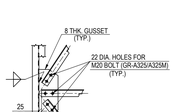
DETAIL - F
(SCALE 1:20)



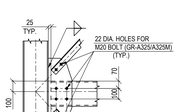
DETAIL - D
(SCALE 1:20)



DETAIL - H
(SCALE 1:20)



DETAIL - G
(SCALE 1:20)



DETAIL - C
(SCALE 1:20)

NOTES:

1. ALL DIMENSIONS & CO-ORDINATES ARE IN MILLIMETER AND LEVELS ARE IN METER UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36,
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
4. ALL WELDS SHALL BE OF SHOP QUALITY & TO BE TESTED AS PER QUALITY ASSURANCE PLAN.
5. ALL PERMANENT STRUCTURAL BOLTS CONFORM TO ASTM A 325M.
6. PAINTING ON STRUCTURES WILL BE AS PER DESIGN BASIS REPORT (DBR).
7. FOR SPLICE DETAIL REFER DRAWING NO. (MD-216-1000-EG-ST-DED-1011)

LEGENDS:-

TOS - TOP OF STEEL
U/S - UNDER SIDE
CL - CENTER LINE
RL - REDUCED LEVEL
T.O.F.F.L - TOP OF FINISHED FLOOR LEVEL
FGL - FINISHED GROUND LEVEL
F/S - FAR SIDE
N/S - NEAR SIDE
DN. - DOWN
W.P. WORKING POINT
GR. - GRATING
TYP. - TYPICAL
PLT. - PLATE
THK. - THICKNESS
STIFF. - STIFFNER
CONN. - CONNECTION

MD-216-1000-EG-ST-CRT-1001	CIVIL,STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA	A01	18/05/2020	ISSUED FOR APPROVAL	MK	SYC	SGI	PRI
DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP EMP

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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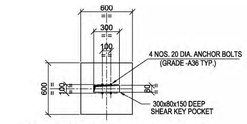
LARSEN & TOUBRO DRAWING NUMBER	O19045_450_S_G_26200_001		REVISION 0
OUTOTEC DRAWING NUMBER	900707__GAU01__HL15__MTL01__33805	DOCUMENT ID OU100674668	REVISION 0
DRAWN	MK	14/05/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SYC	15/05/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	SGI	16/05/2020	ACIDULATION AREA
APPROVED	PRI	18/05/2020	PLAN, ELEVATIONS, SECTIONS & DETAILS
EMPLOYER			
SCALE 1 : 20, 100 & 150	A0	DRAWING NUMBER MD-216-3380-EG-ST-GAD-1001	REV A01

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	18-08-2020	21-10-2020	12-10-2023				Date	18-08-2020	21-10-2020	12-10-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 3 **RELATED CIVIL GA & RC DRAWINGS OF NEARBY STRUCTURE**

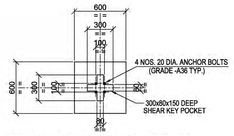
DRAWING NUMBER: MD-216-3130-EG-CV-GAD-1004 | REV: D01



DETAIL OF PEDESTAL MKD. P1

(FOR CONVEYOR MKD. 3130-FDR-003)

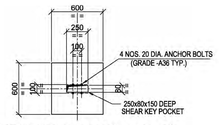
(SCALE-1:25)



DETAIL OF PEDESTAL MKD. P2

(FOR CONVEYOR MKD. 3130-FDR-003)

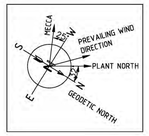
(SCALE-1:25)

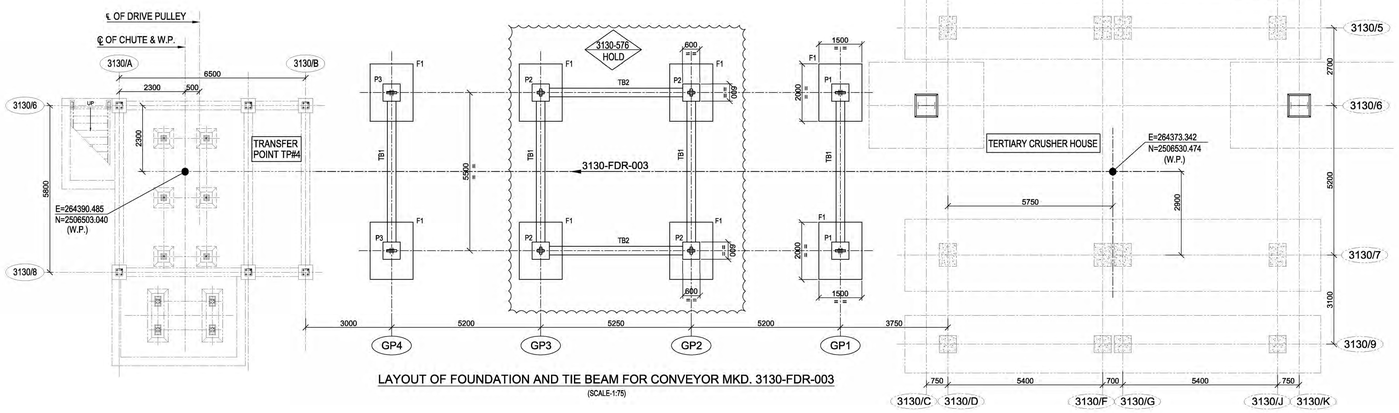


DETAIL OF PEDESTAL MKD. P3

(FOR CONVEYOR MKD. 3130-FDR-003 & 004)

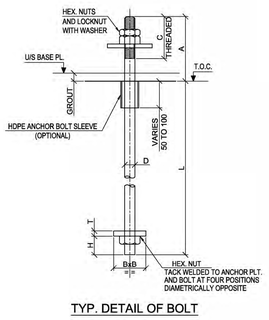
(SCALE-1:25)



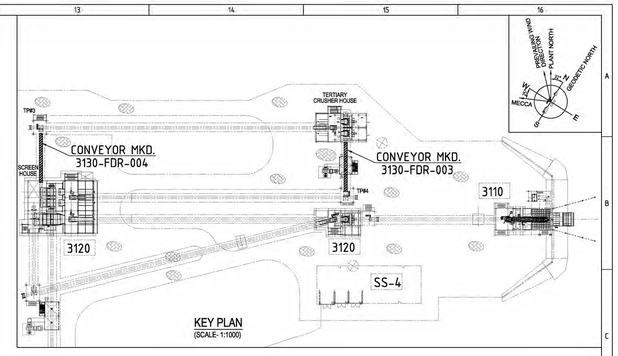


LAYOUT OF FOUNDATION AND TIE BEAM FOR CONVEYOR MKD. 3130-FDR-003

(SCALE-1:75)



TYP. DETAIL OF BOLT

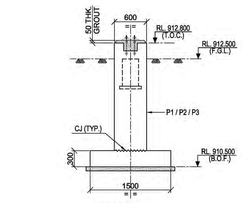


KEY PLAN

(SCALE- 1:1000)

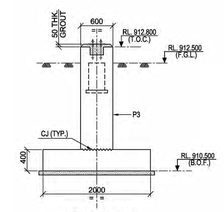
SCHEDULE OF BOLTS						
D BOLT DIA. MM	BxB BOTTOM PLATE (SQUARE) SQ MM	DIMENSIONS (MM)				
		L	A	C	T	H
20	60x60	450	160	130	10	30

SCHEDULE OF BEAM			
BEAM MKD.	SIZE		LEVEL
	WIDTH (mm.)	DEPTH (mm.)	
TB1	300	500	RL. 912.500 (T.O.B.)
TB2	300	400	RL. 912.500 (T.O.B.)
TB3	300	500	RL. 912.500 (T.O.B.)



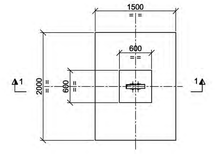
SECTION 1-1

(SCALE-1:40)



SECTION 2-2

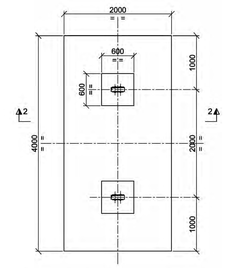
(SCALE-1:40)



DETAIL OF FOUNDATION MKD. F1

(FOR CONVEYOR MKD. 3130-FDR-003)

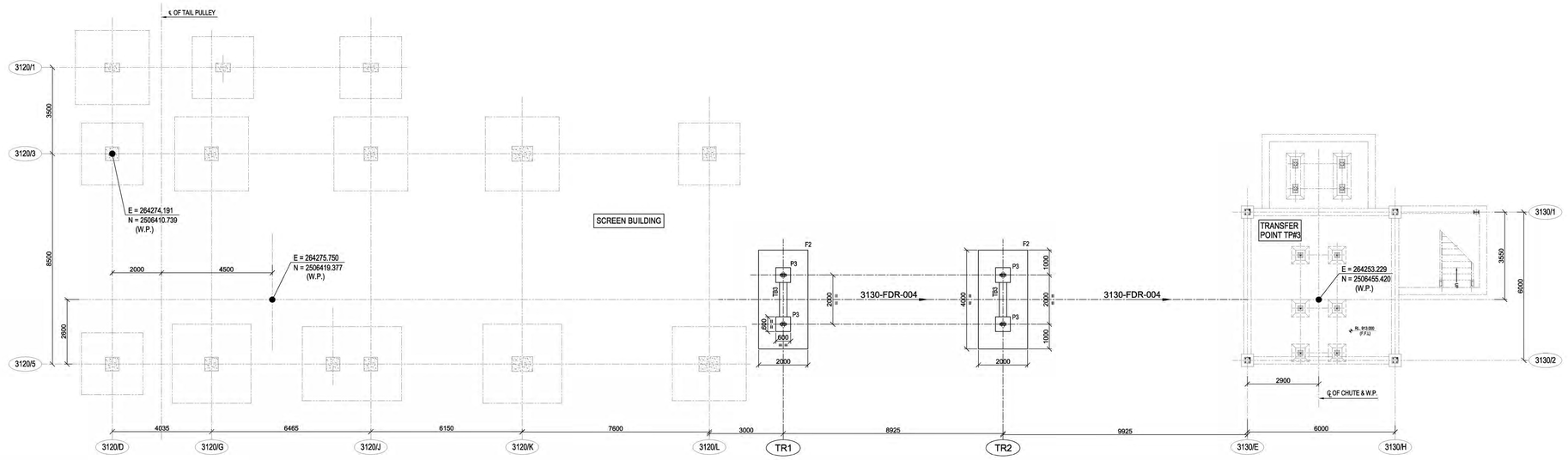
(SCALE-1:40)



DETAIL OF FOUNDATION MKD. F2

(FOR CONVEYOR MKD. 3130-FDR-004)

(SCALE-1:40)



LAYOUT OF FOUNDATION AND TIE BEAM FOR CONVEYOR MKD. 3130-FDR-004

(SCALE-1:75)

LEGEND :-

R.L. - REDUCED LEVEL
℄ - CENTRE LINE
CJ - CONSTRUCTION JOINT
FDN. - FOUNDATION.
T.O.C. - TOP OF CONCRETE.
F.G.L. - FINISHED GROUND LEVEL.
F.F.L. - FINISHED FLOOR LEVEL.
B.O.F. - BOTTOM OF FOUNDATION
TYP. - TYPICAL
B.O.B. - BOTTOM OF BEAM.
B.O.M. - BOTTOM OF MONORAIL.
R.C.C. - REINFORCED CEMENT CONCRETE
P.C.C. - PLAIN CEMENT CONCRETE
IP - INSERT PLATE
MKD. - MARKED.
THK. - THICK

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. GRADE OF CONCRETE SHALL BE C35 FOR FOUNDATION.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615 - GRADE 60 (fy = 420 MPa).
4. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDERED 500KN/ SQ. M.
5. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS
   a) FOUNDATION = 75 mm (BOTTOM & SIDE) 50 mm. (TOP)
   b) COLUMN & PEDESTAL = 50mm
   c) BEAM = 50mm
   d) SLAB = 40mm
6. FOR MINIMUM LAP LENGTH OF REINFORCEMENT AND OTHER GENERAL NOTES FOR CIVIL WORKS REFER STANDARD DRG : MD-216-1000-EG-CV-DED-1001
7. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD - 216 - 3130 -EG -ST - GAD - 1017 TO MD - 216 - 3130 -EG -ST - GAD - 1019.
8. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO. MD - 216 - 1000 -EG -ST - CRT - 1001
9. FOR OTHER STANDARD REINFORCEMENT DETAILS REFER DRG : MD-216-1000-EG-CV-DED-1005

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-ELV-1015	TERTIARY CRUSHING FEEDER 3130-FDR-004-ELEVATION	D01	12/10/2023	AS BUILT	AMJ	SBY	APL	PRI	
MD-216-3130-EG-ME-LPX-1008	TERTIARY CRUSHING FEEDER 3130-FDR-004-PLAN	C01	29/04/2020	ISSUED FOR CONSTRUCTION, REVISED AS PER EMPLOYER	AMJ	SBY	APL	PRI	
MD-216-3110-EG-ME-ELV-1013	TERTIARY CRUSHING FEEDER 3130-FDR-003-ELEVATION			COMMENTS DATED 27/04/2020 & REVISED AS MKD. C01					
MD-216-3130-EG-ME-LPX-1007	TERTIARY CRUSHING FEEDER 3130-FDR-003-PLAN	A01	04/04/2020	ISSUED FOR REFERENCE	AMJ	SBY	APL	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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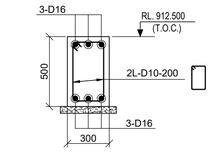
LARSEN & TOUBRO DRAWING NUMBER: D19045_100_C_G_24056_007 | REVISION: 2

OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31360 | DOCUMENT ID: OU100670367 | REVISION: 2

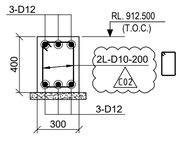
DRAWN	AMJ	02/04/2020
CHECKED	SBY	03/04/2020
ENGINEER	APL	01/04/2020
APPROVED	PRI	04/04/2020
EMPLOYER		
SCALE	1:100,50,20,10	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GA OF CONVEYORS 3130-FDR-003 & FDR-004
LAYOUT & DETAIL

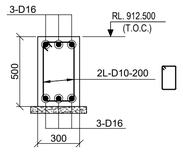
DRAWING NUMBER: MD - 216 - 3130 - EG - CV - GAD - 1004 | REV: D01



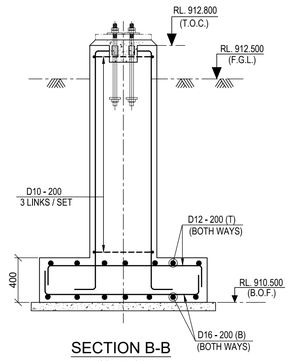
RC DETAIL OF
TIE BEAM MKD. TB1
(SCALE- 1:20)



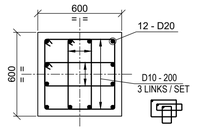
RC DETAIL OF
TIE BEAM MKD. TB2
(SCALE- 1:20)



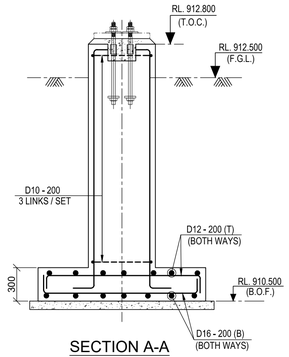
RC DETAIL OF
TIE BEAM MKD. TB3
(SCALE- 1:20)



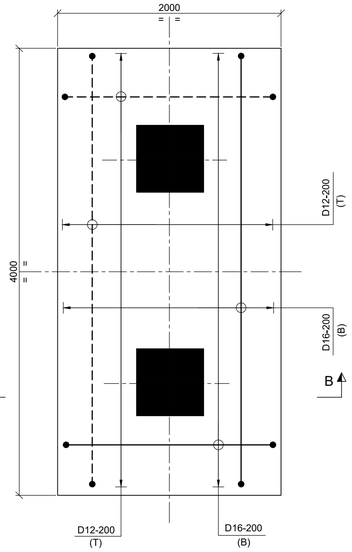
SECTION B-B
(SCALE-1:25)



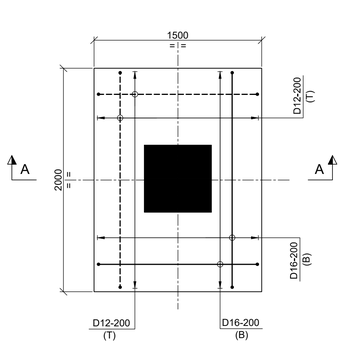
RC DETAIL OF
PEDESTAL MKD. P1, P2 & P3
(P1 - 2 NOS. REQUIRED)
(P2 - 4 NOS. REQUIRED)
(P3 - 6 NOS. REQUIRED)
(SCALE- 1:20)



SECTION A-A
(SCALE-1:25)



R.C. DETAIL OF FOUNDATION MKD. F2
(F2 - 2 NOS. REQUIRED)
(SCALE- 1:25)



R.C. DETAIL OF FOUNDATION MKD. F1
(F1 - 8 NOS. REQUIRED)
(SCALE- 1:25)

LEGENDS :-

R.L. - REDUCED LEVEL
℄ - CENTRE LINE
CJ - CONSTRUCTION JOINT
FDN. - FOUNDATION.
T.O.C. - TOP OF CONCRETE.
F.G.L. - FINISHED GROUND LEVEL.
F.F.L. - FINISHED FLOOR LEVEL.
B.O.F. - BOTTOM OF FOUNDATION
TYP. - TYPICAL
R.C. - REINFORCED CONCRETE
P.C.C - PLAIN CEMENT CONCRETE
MKD. - MARKED.
THK. - THICK
D - DIAMETER OF BAR
B/W - BOTH WAYS
Ld - DEVELOPMENT LENGTH
BOTTOM REINFORCEMENT (B) - ———
TOP REINFORCEMENT (T) - – – – – –
ALTH - ALLTHROUGH
B/F - BOTHFACE
REIN. - REINFORCEMENT

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. GRADE OF CONCRETE SHALL BE C35 FOR FOUNDATION.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615 - GRADE 60 (fy = 420 MPa).
4. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDERED 500KN/ SQ. M.
5. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS
   a) FOUNDATION = 75 mm (BOTTOM & SIDE) 50 mm. (TOP)
   b) COLUMN & PEDESTAL = 50mm
   c) BEAM = 50mm
   d) SLAB = 40mm
6. FOR MINIMUM LAP LENGTH OF REINFORCEMENT AND OTHER GENERAL NOTES FOR CIVIL WORKS REFER STANDARD DRG : MD-216-1000-EG-CV-DED-1001
7. READ THIS DRAWING IN CONJUNCTION WITH DRG: MD-216-3130-EG-CV-GAD-1004
8. FOR OTHER STANDARD REINFORCEMENT DETAILS REFER DRG : MD-216-1000-EG-CV-DED-1005

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D02	12/10/2023	AS BUILT	JKM	SBY	APL	PRI	
		C02	11/08/2020	ISSUED FOR CONSTRUCTION, REVISED AS MKD. C02	JKM	SBY	APL	PRI	
		C01	29/04/2020	ISSUED FOR CONSTRUCTION, REVISED AS PER EMPLOYER	AMJ	SBY	APL	PRI	
				COMMENTS DATED 27/04/2020 & REVISED AS MKD. C01					
MD-216-3130-EG-CV-GAD-1004	GA OF CONVEYORS 3130-FDR-003 & FDR-004	A01	04/04/2020	ISSUED FOR REFERENCE	AMJ	SBY	APL	PRI	

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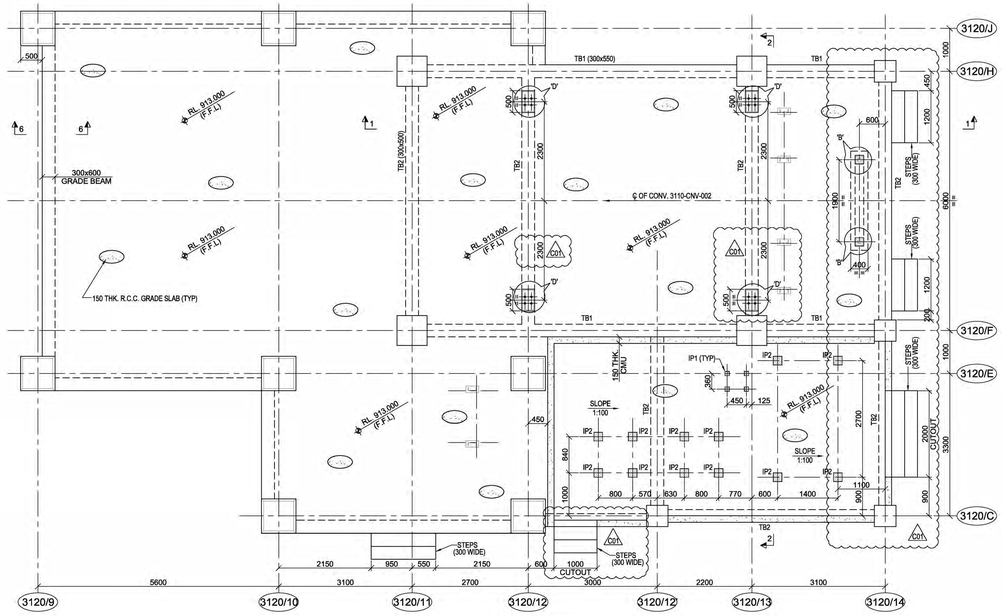
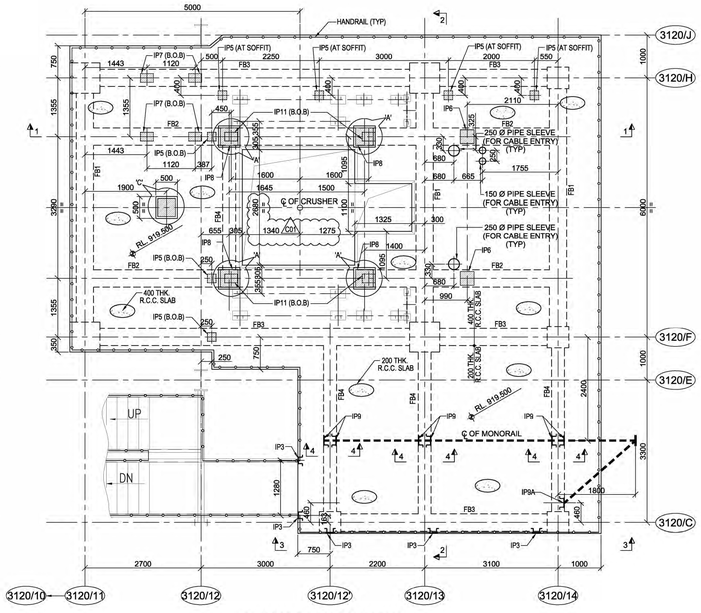
LARSEN & TOUBRO DRAWING NUMBER			REVISION
O19045_100_C_D_24057_007			3
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707__GAU01__ZC01__CTB17__31304		OU100670381	3
DRAWN	AMJ	02/04/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SBY	03/04/2020	DETAIL DRAWING
ENGINEER	APL	01/04/2020	RC DETAILS OF CONVEYORS 3130-FDR-003 & FDR-004
APPROVED	PRI	04/04/2020	RC DETAILS
EMPLOYER			
SCALE 1:100,20,25		A1	DRAWING NUMBER: MD-216-3130-EG-CV-DED-1004 REV: D02

APPROX. CONC. QTY. FOR PLANNING PURPOSE
A) REBAR = 3.3 T.

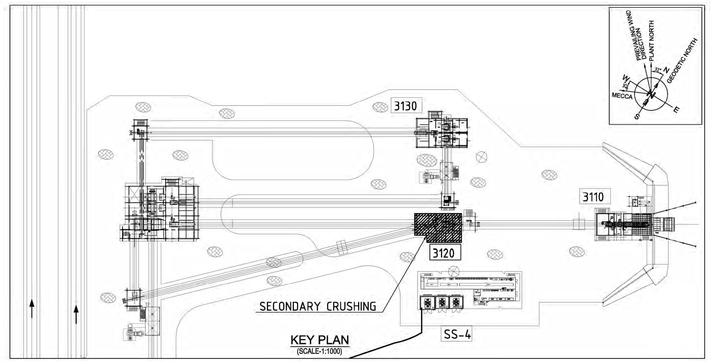
DRAWING NUMBER: MD-216-3120-EG-CV-GAD-1004 | REV: D01



PLAN OF GRADE SLAB AT RL. 913.000 AND TIE BEAM AT RL. 912.850 (T.O.B)
(SCALE-1:50)



PLAN AT RL 919.500M LEVEL
(SCALE-1:50)



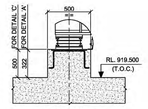
KEY PLAN
(SCALE-1:1000)

SCHEDULE OF BEAM

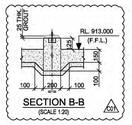
BEAM MKD.	SIZE WIDTH (mm.)	SIZE DEPTH (mm.)	LEVEL
TB1	300	550	RL. 912.850 (T.O.C.)
TB2	300	500	
FB1	400	1000	RL. 919.500 (T.O.C.)
FB2	400	900	
FB3	400	700	
FB4	300	500	

FOUNDATION SCHEDULE

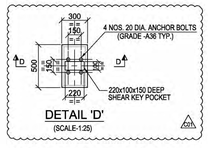
FDN. MKD.	TYPE MKD.	PEDESTAL/COLUMN MKD.	FOUNDATION DETAILS L	B	D	BOTTOM OF FDN.	TOP OF PEDESTAL/COLUMN	NOS. FDN.
F1	I	P1	3000	2000	400	RL. 910.500	RL. 913.450	2
F1A	I	C1	3000	2000	400	RL. 910.500	RL. 919.500	2
F2	I	P3	4500	3300	900	RL. 910.500	RL. 913.450	2
CF1 & CF1A	III	P2 & P3	8300	3500	800	RL. 910.500	RL. 913.450	2
CF2	III	C2	9000	2000	500	RL. 910.500	RL. 919.500	1
CF3	III	C2	3000	6100	600	RL. 910.500	RL. 919.500	2



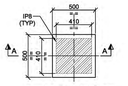
SECTION A-A
(SCALE 1:20)



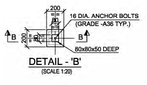
SECTION B-B
(SCALE 1:20)



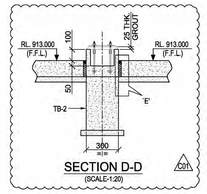
DETAIL 'D'
(SCALE-1:25)



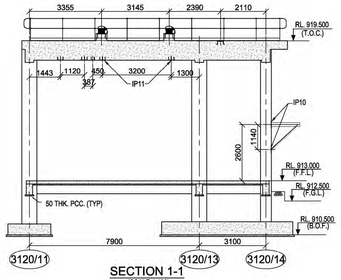
DETAIL 'A' & 'C'
(SCALE 1:20)



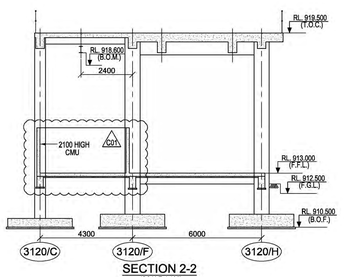
DETAIL - 'B'
(SCALE 1:20)



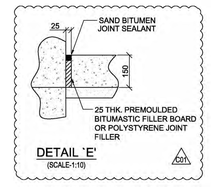
SECTION D-D
(SCALE-1:25)



SECTION 1-1
(SCALE-1:100)



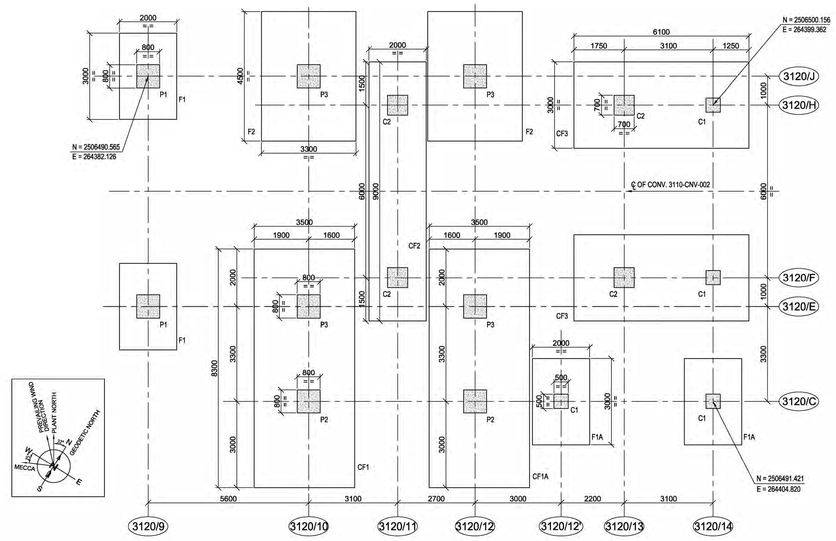
SECTION 2-2
(SCALE-1:100)



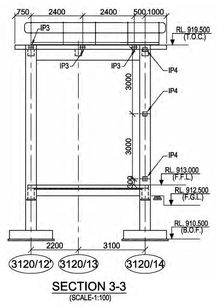
DETAIL 'E'
(SCALE-1:10)

SCHEDULE OF BOLTS

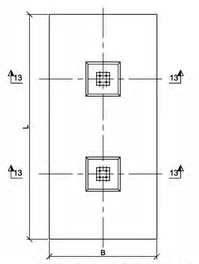
D BOLT DIA. MM	BxB BOTTOM PLATE (SQUARE) SQ MM	DIMENSIONS (MM) L	A	C	T	H	
16	50x50	350	150	100	10	30	C01
20	60x60	450	150	100	10	30	
32	100x100	800	200	150	16	35	
36	110x110	1000	225	175	20	35	



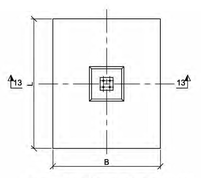
LAYOUT OF FOUNDATION
(SCALE-1:75)



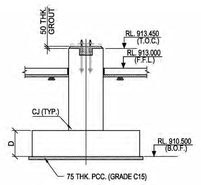
SECTION 3-3
(SCALE-1:100)



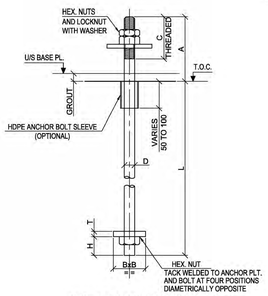
TYPICAL FOUNDATION PLAN
(TYPE-II)
(SCALE-1:50)



TYPICAL FOUNDATION PLAN
(TYPE-II)
(SCALE-1:50)



SECTION 13-13
(SCALE-1:50)



TYP. DETAIL OF BOLT

LEGENDS :-

R.L - REDUCED LEVEL
℄ - CENTRE LINE
CJ - CONSTRUCTION JOINT
FDN. - FOUNDATION.
T.O.C. - TOP OF CONCRETE.
F.G.L. - FINISHED GROUND LEVEL.
F.F.L. - FINISHED FLOOR LEVEL.
B.O.F. - BOTTOM OF FOUNDATION
TYP. - TYPICAL.
B.O.B. - BOTTOM OF BEAM.
B.O.M. - BOTTOM OF MONORAIL.
R.C.C - REINFORCED CEMENT CONCRETE
P.C.C - PLAIN CEMENT CONCRETE
IP. - INSERT PLATE
MKD. - MARKED.
THK. - THICK
CMU - CONCRETE MASONRY UNIT (C01)

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. GRADE OF CONCRETE SHALL BE C35 FOR FOUNDATION.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615 - GRADE 60 (fy = 420 MPa).
4. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDERED 500KN/SQ. M.
5. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS
   a) FOUNDATION = 75 mm (BOTTOM & SIDE) 50 mm. (TOP)
   b) COLUMN & PEDESTAL = 50mm
   c) BEAM = 50mm
   d) SLAB = 40mm
6. MINIMUM LAP LENGTH OF REINFORCEMENT SHALL BE 40 x DIA. OF BAR.
7. THIS DRG. SHALL BE READ IN CONJUNCTION WITH DRG. NO. MD-216-3120-EG-CV-GAD-1007 & MD-216-3120-EG-CV-DED-1004 (C01)
8. FOR DETAIL OF HANDRAIL FIXING REFER STANDARD DRAWING. (C01)

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3120-EG-ME-ELV-1012	SECONDARY CRUSHER HOUSE-ELEVATION								
MD-216-3120-EG-ME-ELV-1013	SECONDARY CRUSHER HOUSE-ELEVATION								
MD-216-3120-EG-ME-LPX-1014	SECONDARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1015	SECONDARY CRUSHER HOUSE-PLAN VIEW	D01	12/10/2023	AS BUILT	AMJ	SBY	APL	PRI	
MD-216-3120-EG-ME-LPX-1016	SECONDARY CRUSHER HOUSE-PLAN VIEW	C01	11/03/2020	ISSUED FOR CONSTRUCTION, REVISED AS PER EMPLOYER COMMENTS DATED 26.02.2020 & REVISED AS MKD. (C01)	HSK	SBY	APR	AG	
MD-216-3120-EG-ME-LPX-1017	SECONDARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3120-EG-ME-LPX-1018	SECONDARY CRUSHER HOUSE-PLAN VIEW	A01	27/01/2020	ISSUED FOR REFERENCE	JKM	SBY	APR	AG	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: 019045_100_C_G_24062_001 — REVISION: 2
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31252 — DOCUMENT ID: OU100670443 — REVISION: 2

DRAWN	JKM	10/01/2020
CHECKED	SBY	21/01/2020
ENGINEER	APR	20/01/2020
APPROVED	AG	27/01/2020
EMPLOYER		

SCALE: 1:100,1:75,1:50,1:25,1:20,1:10 — A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
SECONDARY CRUSHING
LAYOUT & DETAIL

DRAWING NUMBER: MD-216-3120-EG-CV-GAD-1004 — REV: D01

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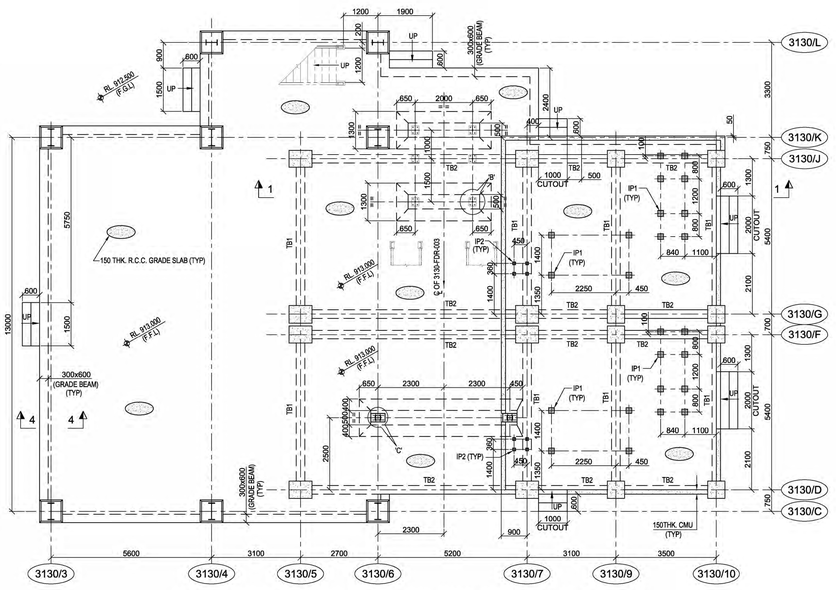
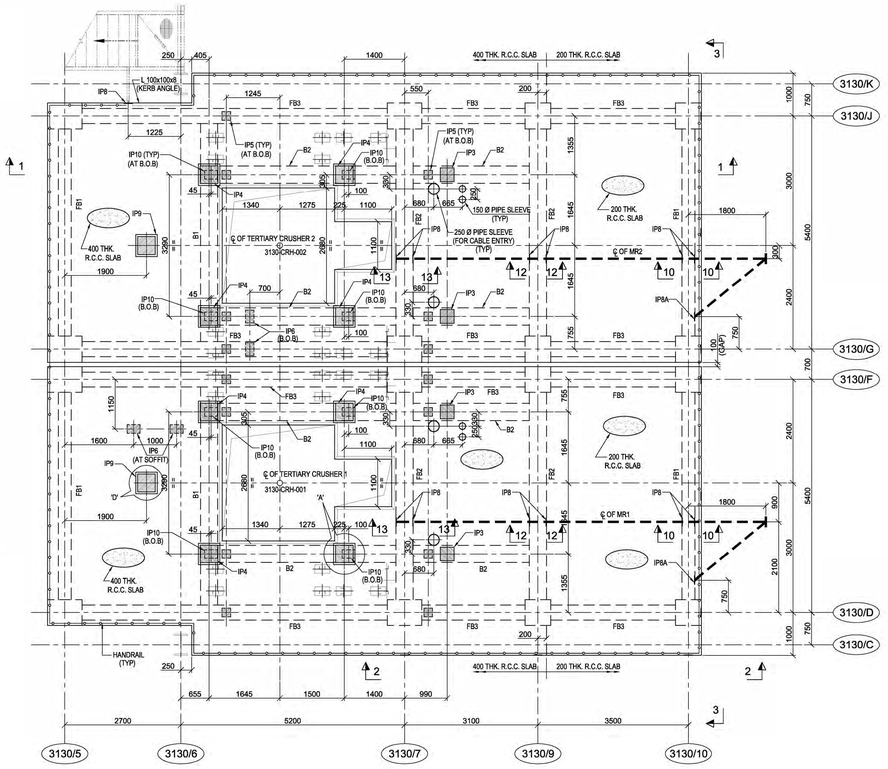


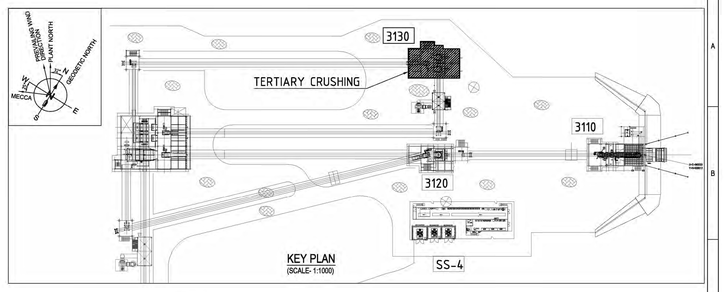
PLAN OF GRADE SLAB AT RL.913.000 (F.F.L) AND TIE BEAM AT RL. 912.850 (T.O.B)

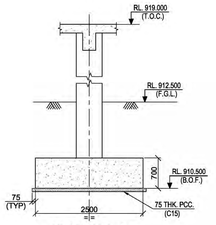
(SCALE - 1:75)



PLAN AT RL. 919.000 (T.O.C)

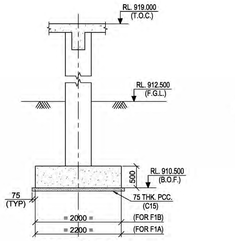
(SCALE - 1:50)





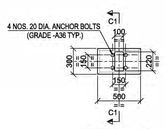
SECTION A-A

(SCALE - 1:50)



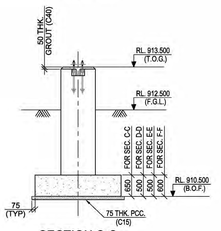
SECTION B-B

(SCALE - 1:50)



DETAIL - 'C'

(SCALE-1:25)

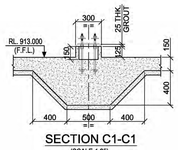


SECTION C-C
SECTION D-D
SECTION E-E
SECTION F-F

(SCALE - 1:50)

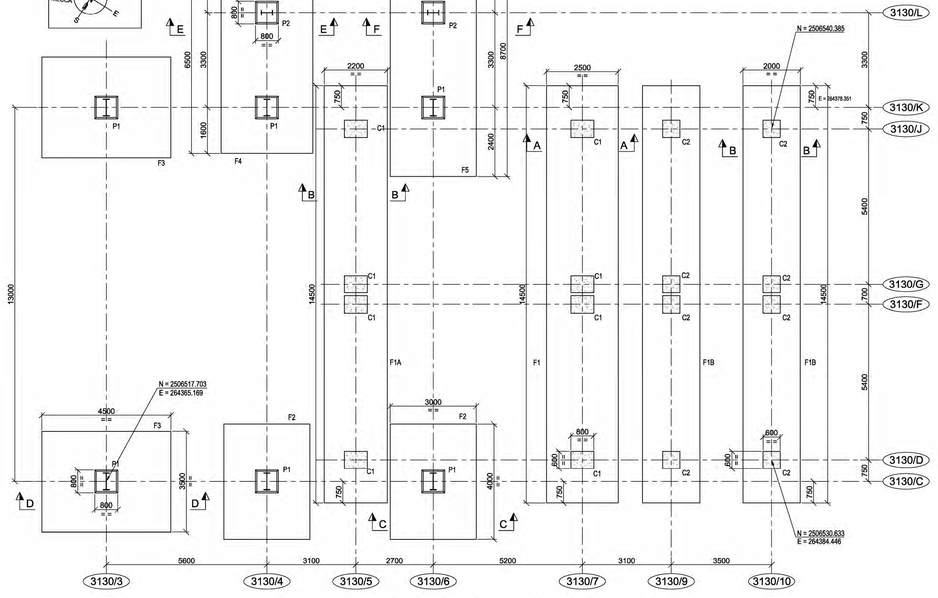
SCHEDULE OF BEAM

BEAM MKD.	SIZE WIDTH (mm.)	SIZE DEPTH (mm.)	LEVEL
TB1	300	500	RL.+912.850 (T.O.B.)
TB2	300	550	
FB1	300	600	
FB2	400	900	
FB3	350	1100	RL.+919.000 (T.O.C.)
B1	400	900	
B2	400	800	



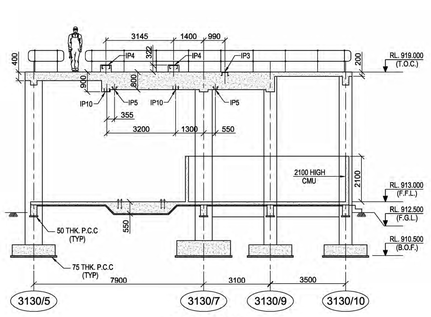
SECTION C1-C1

(SCALE-1:25)



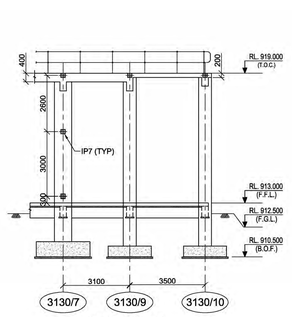
LAYOUT OF FOUNDATION AT RL. 910.500 (B.O.F)

(SCALE - 1:75)



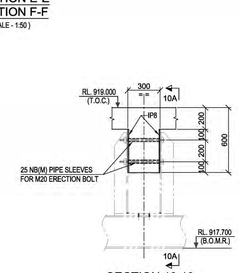
SECTION 1-1

(SCALE - 1:100)



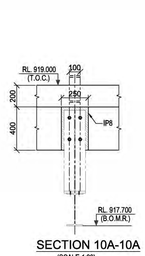
SECTION 2-2

(SCALE - 1:100)



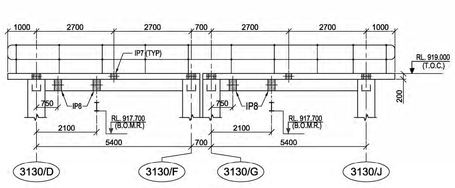
SECTION 10-10

(SCALE-1:20)



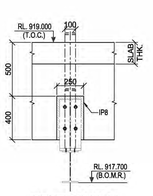
SECTION 10A-10A

(SCALE-1:20)



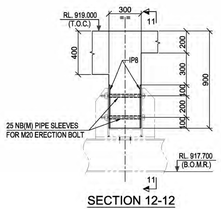
SECTION 3-3

(SCALE - 1:75)



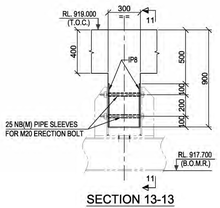
SECTION 11-11

(SCALE-1:20)



SECTION 12-12

(SCALE-1:20)



SECTION 13-13

(SCALE-1:20)

LEGENDS :-

RL. - REDUCED LEVEL
℄ - CENTRE LINE
T.O.C. - TOP OF CONCRETE.
F.G.L. - FINISHED GROUND LEVEL.
B.O.F. - BOTTOM OF FOUNDATION
B.O.B. - BOTTOM OF BEAM
TYP. - TYPICAL.
R.C.C. - REINFORCED CEMENT CONCRETE
P.C.C - PLAIN CEMENT CONCRETE
MKD. - MARKED
THK. - THICK
MR. - MONORAIL
B.O.M.R. - BOTTOM OF MONORAIL
SEC. - SECTION
IP - INSERT PLATE
MM - MILLIMETER
SQ. - SQUARE
CMU - CONCRETE MASONRY UNIT

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. GRADE OF CONCRETE SHALL BE C35 FOR FOUNDATION.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615 - GRADE 60 (fy = 420 MPa).
4. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDERED 500KN/SQ. M.
5. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS
   a) FOUNDATION = 75mm (BOTTOM & SIDE) 50 mm (TOP)
   b) TIE BEAM = 50 mm c) COLUMN = 50mm d) BEAM = 50mm e) SLAB = 40mm
6. FOR MINIMUM LAP LENGTH OF REINFORCEMENT REFER STANDARD DRAWING NO. MD-216-1000-EG-CV-DED-1001
7. THIS DRG. SHALL BE READ IN CONJUNCTION WITH DRG. NO. MD-216-3130-EG-CV-GAD-1008 & MD-216-3130-EG-CV-DED-1007
8. FOR DETAIL OF HANDRAIL FIXING REFER STANDARD DRAWING.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-LPX-1009	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1010	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1012	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1013	TERTIARY CRUSHER HOUSE-PLAN VIEW	C01	09/04/2020	ISSUED FOR CONSTRUCTION, REVISED AS PER EMPLOYER COMMENTS DATED 04/04/2020 & REVISED AS MKD	HSK	SBY	RPK	AG	
MD-216-3130-EG-ME-LPX-1014	TERTIARY CRUSHER HOUSE-PLAN VIEW	A02	12/03/2020	REVISED AS PER EMPLOYER COMMENTS ON 26/02/2020	HSK	SBY	RPK	AG	
MD-216-3130-EG-ME-ELV-1017	TERTIARY CRUSHER HOUSE-ELEVATION	A01	27/01/2020	ISSUED OF REFERENCE	HSK	SBY	RPK	AG	
MD-216-3130-EG-ME-ELV-1018	TERTIARY CRUSHER HOUSE-ELEVATION								

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER	019045_100_C_G_24065_001
REVISION	01
OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31372
DOCUMENT ID	OU1006704544
REVISION	2

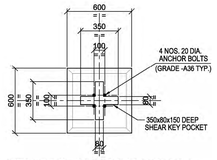
DRAWN	HSK	10/01/2020
CHECKED	SBY	21/01/2020
ENGINEER	RPK	20/01/2020
APPROVED	AG	27/01/2020
EMPLOYER		

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
LAYOUT & DETAIL

SCALE: 1:100,1:75,1:50,1:25,1:20,1:10 — A0

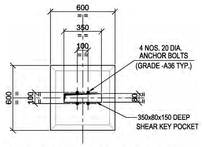
DRAWING NUMBER: MD - 216 - 3130 - EG - CV - GAD - 1007 — REV: C01

DRAWING NUMBER: MD-216-3130-EG-CV-GAD-1006 | REV: D01



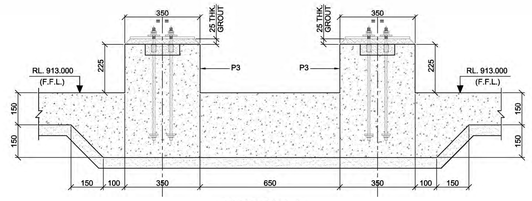
DETAIL OF PEDESTAL MKD. P1

(SCALE-1:20)



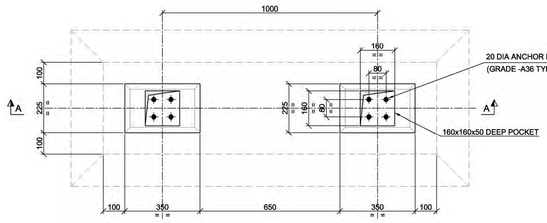
DETAIL OF PEDESTAL MKD. P2

(SCALE-1:20)



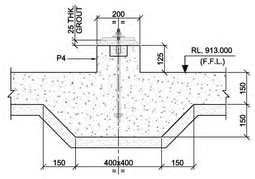
SECTION A-A

(SCALE 1:10)



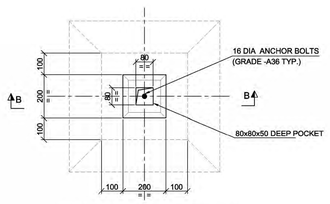
DETAIL OF PEDESTAL MKD. P3

(SCALE 1:10)



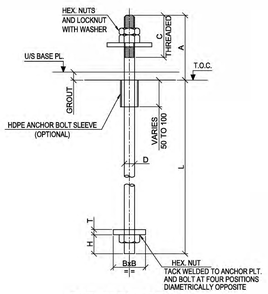
SECTION B-B

(SCALE 1:10)

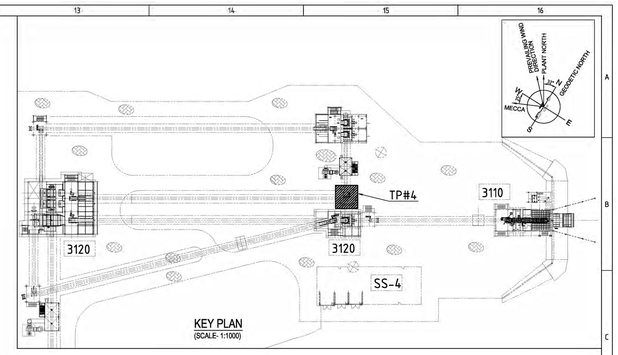


DETAIL OF PEDESTAL MKD. P4

(SCALE 1:10)



TYP. DETAIL OF BOLT

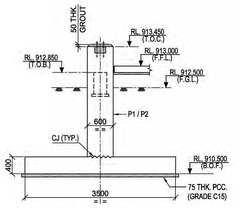


KEY PLAN

(SCALE- 1:1000)

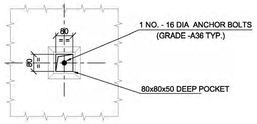
SCHEDULE OF BOLTS

D BOLT DIA. MM	BxB BOTTOM PLATE (SQUARE) SQ.MM	DIMENSIONS (MM)				
		L	A	C	T	H
16	50x50	350	120	100	10	30
20	60x60	450	160	130	10	30



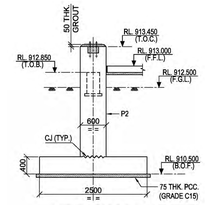
SECTION 2-2

(SCALE-1:50)



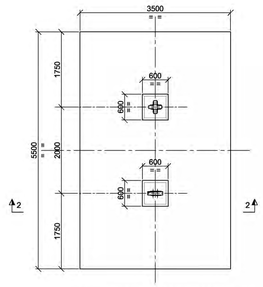
VIEW X-X

(SCALE 1:10)



SECTION 1-1

(SCALE-1:50)

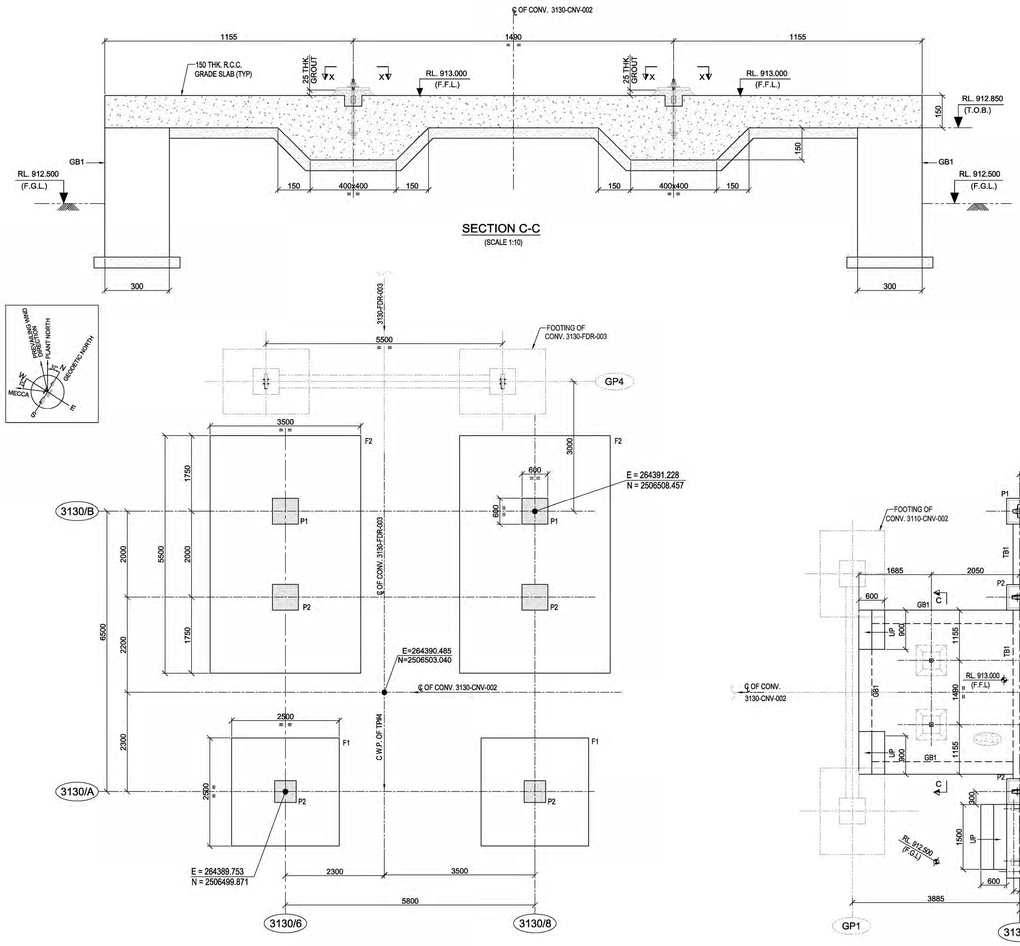


DETAIL OF FOUNDATION MKD. F2

(SCALE-1:50)

SCHEDULE OF BEAM

BEAM MKD.	SIZE		LEVEL
	WIDTH (mm.)	DEPTH (mm.)	
TB1	300	500	RL. 912.850 (T.O.B.)
GB1	300	600	RL. 912.850 (T.O.B.)

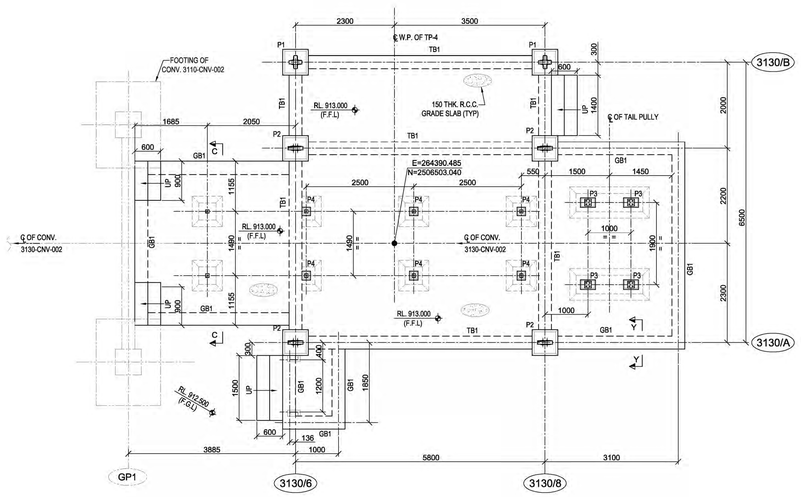


SECTION C-C

(SCALE 1:10)

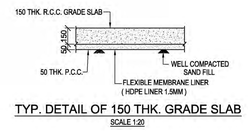
LAYOUT OF FOUNDATION AT RL. 910.500 (B.O.F)

(SCALE-1:50)



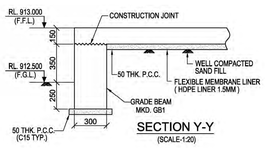
PLAN OF GRADE SLAB AT RL. 913.000 (F.F.L)

(SCALE-1:50)



TYP. DETAIL OF 150 THK. GRADE SLAB

SCALE 1:20



SECTION Y-Y

(SCALE-1:20)

LEGENDS :-

R.L. - REDUCED LEVEL
℄ - CENTRE LINE
CJ - CONSTRUCTION JOINT
FDN. - FOUNDATION.
T.O.C. - TOP OF CONCRETE.
F.G.L. - FINISHED GROUND LEVEL.
F.F.L. - FINISHED FLOOR LEVEL.
B.O.F. - BOTTOM OF FOUNDATION
TYP. - TYPICAL.
B.O.B. - BOTTOM OF BEAM.
B.O.M. - BOTTOM OF MONORAIL.
R.C.C - REINFORCED CEMENT CONCRETE
P.C.C - PLAIN CEMENT CONCRETE
IP - INSERT PLATE
MKD. - MARKED.
THK. - THICK

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. GRADE OF CONCRETE SHALL BE C35 FOR FOUNDATION.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615 - GRADE 60 (fy = 420 MPa).
4. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDERED 500KN/ SQ. M.
5. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS
   a) FOUNDATION = 75 mm (BOTTOM & SIDE) 50 mm. (TOP)
   b) COLUMN & PEDESTAL = 50mm
   c) BEAM = 50mm
   d) SLAB = 40mm
6. FOR MINIMUM LAP LENGTH OF REINFORCEMENT REFER STANDARD DRG : MD-216-1000-EG-CV-DED-1001
7. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD - 216 - 3130 -EG -ST - GAD - 1026 & MD - 216 - 3130 -EG -ST - GAD - 1027
8. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO. MD - 216 - 1000 -EG -ST - CRT - 1001
9. FOR OTHER STANDARD REINFORCEMENT DETAILS REFER DRG : MD-216-1000-EG-CV-DED-1005

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	12/10/2023	AS BUILT	AMJ	SBY	APL	PRI	
		C01	11/05/2020	ISSUED FOR CONSTRUCTION, REVISED AS PER EMPLOYER COMMENTS DATED 05/05/2020 & REVISED AS MKD. C01	AMJ	SBY	RPK	PRI	
MD-216-3130-EG-ME-ELV-1021	TERTIARY CRUSHING TRANSFER POINT TP#4 - ELEVATION								
MD-216-3130-EG-ME-LPX-1016	TERTIARY CRUSHING TRANSFER POINT TP#4 - PLAN VIEW	A01	10/04/2020	ISSUED FOR REFERENCE	AMJ	SBY	RPK	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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Saudi Arabian Mining Company

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LARSEN & TOUBRO DRAWING NUMBER: 019045_100_C_G_24108_002 | REVISION: 2
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31302 | DOCUMENT ID: OU100668046 | REVISION: 2

DRAWN	AMJ	08/04/2020
CHECKED	SBY	09/04/2020
ENGINEER	RPK	07/04/2020
APPROVED	PRI	10/04/2020
EMPLOYER		
SCALE	1:100,50,25,20,10	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
GA OF TP - 4 (3130 CNV-002)
LAYOUT & DETAIL

DRAWING NUMBER: MD - 216 - 3130 - EG - CV - GAD - 1006 | REV: D01

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	18-08-2020	21-10-2020	12-10-2023				Date	18-08-2020	21-10-2020	12-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 4 **STAAD INPUT OF TOTAL FOUNDATION ARRANGEMENT**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 06-Aug-20
JOB PART BAG FILTER & DUCT SUPPORT FOUNDATION
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 13.6725 -3.3 -1.2525; 2 14.0725 -3.3 -1.2525; 3 14.5725 -3.3 -1.2525;
4 15.0725 -3.3 -1.2525; 5 15.4725 -3.3 -1.2525; 6 13.6725 -3.3 -0.8525;
7 14.0725 -3.3 -0.8525; 8 14.5725 -3.3 -0.8525; 9 15.0725 -3.3 -0.8525;
10 15.4725 -3.3 -0.8525; 11 13.6725 -3.3 -0.3525; 12 14.0725 -3.3 -0.3525;
13 14.5725 -3.3 -0.3525; 14 15.0725 -3.3 -0.3525; 15 15.4725 -3.3 -0.3525;
16 13.6725 -3.3 0.1475; 17 14.0725 -3.3 0.1475; 18 14.5725 -3.3 0.1475;
19 15.0725 -3.3 0.1475; 20 15.4725 -3.3 0.1475; 21 13.6725 -3.3 0.5475;
22 14.0725 -3.3 0.5475; 23 14.5725 -3.3 0.5475; 24 15.0725 -3.3 0.5475;
25 15.4725 -3.3 0.5475; 26 13.6725 -3.3 1.7475; 27 14.0725 -3.3 1.7475;
28 14.5725 -3.3 1.7475; 29 15.0725 -3.3 1.7475; 30 15.4725 -3.3 1.7475;
31 8.6725 -3.3 1.7475; 32 9.0725 -3.3 1.7475; 33 9.5725 -3.3 1.7475;
34 10.0725 -3.3 1.7475; 35 10.4725 -3.3 1.7475; 36 13.6725 -3.3 2.1475;
37 14.0725 -3.3 2.1475; 38 14.5725 -3.3 2.1475; 39 15.0725 -3.3 2.1475;
40 15.4725 -3.3 2.1475; 41 8.6725 -3.3 2.1725; 42 9.0725 -3.3 2.1725;
43 9.5725 -3.3 2.1725; 44 10.0725 -3.3 2.1725; 45 10.4725 -3.3 2.1725;
46 8.6725 -3.3 2.6475; 47 9.0725 -3.3 2.6475; 48 9.5725 -3.3 2.6475;
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52 14.0725 -3.3 2.6475; 53 14.5725 -3.3 2.6475; 54 15.0725 -3.3 2.6475;
55 15.4725 -3.3 2.6475; 56 8.6725 -3.3 3.1225; 57 9.0725 -3.3 3.1225;
58 9.5725 -3.3 3.1225; 59 10.0725 -3.3 3.1225; 60 10.4725 -3.3 3.1225;
61 13.6725 -3.3 3.1475; 62 14.0725 -3.3 3.1475; 63 14.5725 -3.3 3.1475;
64 15.0725 -3.3 3.1475; 65 15.4725 -3.3 3.1475; 66 8.6725 -3.3 3.5475;
67 9.0725 -3.3 3.5475; 68 9.5725 -3.3 3.5475; 69 10.0725 -3.3 3.5475;
70 10.4725 -3.3 3.5475; 71 13.6725 -3.3 3.5475; 72 14.0725 -3.3 3.5475;
73 14.5725 -3.3 3.5475; 74 15.0725 -3.3 3.5475; 75 15.4725 -3.3 3.5475;
76 -0.14 -3.2 -3.365; 77 0.7225 -3.2 -3.365; 78 1.285 -3.2 -3.365;
79 1.91 -3.2 -3.365; 80 2.66 -3.2 -3.365; 81 3.16 -3.2 -3.365;
82 3.66 -3.2 -3.365; 83 7.16 -3.2 -3.365; 84 7.66 -3.2 -3.365;
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88 10.0975 -3.2 -3.365; 89 10.96 -3.2 -3.365; 90 -0.14 -3.2 -2.59;
91 0.7225 -3.2 -2.59; 92 1.285 -3.2 -2.59; 93 1.91 -3.2 -2.59;
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134 1.2475 -3.2 -0.3525; 135 1.835 -3.2 -0.3525; 136 2.66 -3.2 -0.3525;
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149 1.76 -3.2 0.385; 150 2.21 -3.2 0.385; 151 2.66 -3.2 0.385;
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161 10.96 -3.2 0.385; 162 -0.14 -3.2 1.01; 163 0.685 -3.2 1.01;
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182 2.21 -3.2 1.635; 183 2.66 -3.2 1.635; 184 3.16 -3.2 1.635;
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197 1.36 -3.2 4.285; 198 1.76 -3.2 4.285; 199 2.51 -3.2 4.285;
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407 -1.55 -0.9 18.385; 408 -1.05 -0.9 18.385; 409 -0.55 -0.9 18.385;
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MEMBER INCIDENCES
1 93 440; 2 440 460; 3 100 441; 4 441 461; 5 121 442; 6 442 462; 7 128 443;
8 443 463; 9 143 448; 10 448 464; 11 13 449; 12 449 465; 13 151 432;
14 432 466; 15 156 433; 16 433 467; 17 48 450; 18 450 468; 19 53 451;
20 451 469; 21 237 435; 22 435 470; 23 238 436; 24 436 471; 25 273 444;
26 444 472; 27 280 445; 28 445 473; 29 303 446; 30 446 475; 31 310 447;
32 447 476; 36 439 474; 37 149 431; 38 431 485; 39 158 434; 40 434 486;
41 243 437; 42 437 487; 43 250 438; 44 438 488; 45 300 439; 46 344 452;
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ELEMENT INCIDENCES SHELL
501 302 303 288 287; 502 287 288 273 272; 503 303 304 289 288;
504 288 289 274 273; 505 272 273 258 257; 506 257 258 243 242;
507 273 274 259 258; 508 258 259 244 243; 509 309 310 295 294;
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556 100 101 87 86; 557 121 122 108 107; 558 107 108 94 93; 559 149 151 136 135;
560 135 136 122 121; 561 156 158 142 141; 562 141 142 128 127;
563 127 128 114 113; 564 113 114 100 99; 565 307 308 293 292;
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663 178 179 163 162; 664 162 163 147 146; 665 179 180 164 163;
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790 413 414 409 408; 791 419 420 415 414; 792 414 415 410 409;
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848 9 14 15 10; 849 13 18 19 14; 850 18 23 24 19; 851 14 19 20 15;
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880 58 68 69 59; 881 34 44 45 35; 882 44 49 50 45; 883 49 59 60 50;
884 59 69 70 60;
START GROUP DEFINITION
ELEMENT
_MX 535 TO 550 552 TO 555 565 TO 573 575 576 581 TO 588 590 TO 592 -
594 TO 596 601 602 604 606 608 TO 650 653 TO 660 663 TO 670 679 TO 694 761 -
762 TO 764 769 TO 772 781 782 789 790 797 TO 799 801 TO 803 811 812 819 820 -
837 838 841 842 847 848 851 TO 854 859 TO 862 867 868
_MY 521 522 529 530 533 534 537 TO 542 545 TO 558 562 TO 570 575 TO 586 591 -
592 TO 596 602 604 606 608 609 612 615 TO 630 635 636 639 TO 646 649 650 653 -
655 657 659 661 663 665 667 669 675 677 679 TO 682 687 TO 694 751 754 755 -
758 759 761 763 765 768 770 772 774 775 778 779 781 783 788 790 792 -
795 TO 804 837 839 842 844 845 847 850 852 853 855 857 859 862 864 866 868
_SHEAR 538 TO 542 545 TO 550 552 TO 555 565 TO 571 573 575 576 578 -
580 TO 584 586 591 592 594 TO 596 601 602 604 615 617 TO 622 624 TO 628 630 -
632 634 TO 636 640 TO 646 648 TO 650 655 657 659 661 663 667 669 675 677 -
679 TO 682 687 TO 694 764 771 781 790 798 802 837 842 847 852 853 859 862 -
868
_MX_600 535 TO 543 545 TO 550 552 TO 555 565 TO 571 573 575 576 581 TO 588 -
590 TO 592 594 TO 596 601 602 604 606 608 TO 636 638 640 TO 650 653 TO 659 -
663 TO 670 679 TO 688 691 692
_MX_400 761 TO 764 769 TO 772 781 782 789 790 797 TO 799 801 TO 803 811 812 -
819 820 837 838 841 842 847 848 851 TO 854 859 TO 862 867 TO 872 881 TO 884
_MY_600 578 580 595 596 598 600 602 604 624 649 650 653 655 657 659 661 663 -
665 667 669 675 677 679 680
_MY_400 751 754 755 758 759 761 763 765 768 770 772 774 775 778 779 781 783 -
788 790 792 795 TO 804 837 839 842 844 845 847 850 852 853 855 857 859 862 -
864 866 868 869 872 873 876 877 880 881 884
_SHEAR_600 538 TO 542 545 TO 550 552 TO 555 565 TO 570 575 576 578 -
580 TO 584 586 591 592 594 TO 596 602 604 615 617 TO 622 624 TO 628 630 632 -
634 TO 636 640 TO 646 648 TO 650 655 657 659 663 669 679 TO 681 687 688 691 -
692
_SHEAR_400 764 771 781 790 798 802 837 TO 839 841 842 844 845 847 848 850 -
851 TO 855 857 859 TO 862 864 866 TO 873 876 877 880 TO 884
END GROUP DEFINITION
ELEMENT PROPERTY
501 TO 604 606 608 TO 694 THICKNESS 0.6
751 TO 784 787 TO 792 795 TO 824 837 TO 884 THICKNESS 0.4
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
```

```
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 35000
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
37 TO 44 PRIS YD 1 ZD 0.9
13 TO 16 21 TO 24 PRIS YD 0.6 ZD 0.6
1 TO 8 25 TO 32 PRIS YD 0.7 ZD 0.5
9 TO 12 17 TO 20 PRIS YD 0.55 ZD 0.55
36 45 TO 61 PRIS YD 0.45 ZD 0.45
*
71 73 TO 75 78 79 82 83 86 TO 90 PRIS YD 0.6 ZD 0.3
97 TO 102 PRIS YD 0.5 ZD 0.3
76 77 80 81 94 95 PRIS YD 0.4 ZD 0.3
91 TO 93 96 PRIS YD 0.3 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
1 TO 430 ELASTIC MAT DIRECT Y SUBGRADE 30000
MEMBER OFFSET
99 START 0.125 0 0
99 END 0.125 0 0
LOAD 1 LOADTYPE Dead  TITLE DEAD LOAD
JOINT LOAD
****1001 to 1004 TR-4 of Secondary Crusher
472 FX 0 FY -31 FZ -0.2
473 FX 0.1 FY -35 FZ -0.1
475 FX 0 FY -59 FZ -12
476 FX 0.1 FY -35 FZ 0
****1005 to 1005 TR-3 of Secondary Crusher
474 FX -0.1 FY -26 FZ -13
477 FX -0.1 FY -45 FZ 13
478 FX 0.1 FY -46 FZ 14
479 FX 0 FY -26 FZ 0
480 FX 0 FY -26 FZ 0
481 FX 0 FY -6 FZ 0
482 FX 0 FY -6 FZ 0
483 FX 0 FY -6 FZ 0
484 FX 0 FY -6 FZ 0
****2001 to 2004 Bag filter
485 FX -12 FY -635 FZ 0
486 FX 12 FY -653 FZ 0
487 FX -12 FY -636 FZ 0
488 FX 12 FY -654 FZ 0
****3001 to 3004 TR-5 of Tertiary Crusher
460 FX -0.1 FY -37 FZ 0
461 FX 0.1 FY -37 FZ 0
462 FX -0.1 FY -37 FZ 0
463 FX 0.1 FY -37 FZ 0
****3011 to 3014 Stair case
464 FX -0.5 FY -49.4 FZ -3.6
```

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465 FX 0.4 FY -43.8 FZ 0
469 FX 0.5 FY -44.6 FZ 0.1
468 FX -0.4 FY -48.7 FZ 3.6
****5001 to 5004 3130-Feeder
466 FX 0.7 FY -21.9 FZ -14.5
467 FX 2.4 FY -47.8 FZ 19.1
470 FX -1.7 FY -24.5 FZ -2.5
471 FX -3.5 FY -49.2 FZ 5
SELFWEIGHT Y -1
ELEMENT LOAD
795 TO 824 PR GY -20.2
775 TO 784 787 TO 792 PR GY -15.7
751 TO 774 PR GY -32.5
501 TO 604 606 608 TO 678 681 TO 688 693 694 PR GY -67
837 TO 884 PR GY -67
*********************************************************************************
LOAD 2 LOADTYPE Live  TITLE LIVE LOAD
JOINT LOAD
472 FX -0.1 FY 0 FZ 0.3
473 FX 0.2 FY 0 FZ 0
475 FX -0.3 FY 0 FZ 0
476 FX 0.2 FY 0 FZ 0
466 FX 1.2 FY -31.6 FZ -27.5
467 FX 1.2 FY -31.6 FZ 27.5
470 FX -1.9 FY -33.2 FZ -3.9
471 FX -1.9 FY -33.2 FZ 3.9
464 FX -0.1 FY -88.8 FZ -8.5
465 FX -0.1 FY -73 FZ -0.1
469 FX 0.1 FY -74.1 FZ 0.1
468 FX 0.1 FY -87.7 FZ 8.4
LOAD 3 LOADTYPE Wind  TITLE WIND LOAD (+X DIR)
JOINT LOAD
472 FX 2.1 FY -16 FZ -2.2
473 FX 2.2 FY 6.1 FZ 2
475 FX 27 FY -127.6 FZ 1.6
476 FX 2.2 FY -36.4 FZ 2
474 FX 29 FY 177.1 FZ -4.9
477 FX 11.7 FY 51.2 FZ -5.9
478 FX 11.7 FY -54.4 FZ 7.5
479 FX 3.2 FY 19.7 FZ 0
480 FX 3.3 FY -19.7 FZ 0
481 FX 2.2 FY 5.1 FZ 0
482 FX 2.2 FY -5.1 FZ 0
483 FX 2.2 FY 5.1 FZ 0
484 FX 2.2 FY -5.1 FZ 0
485 FX 138 FY 388 FZ 25
486 FX 140 FY -388 FZ 0
487 FX 143 FY 398 FZ -19
488 FX 145 FY -398 FZ -6
460 FX 12.5 FY 34.4 FZ 0
461 FX 12.5 FY -34.4 FZ 0
462 FX 12.5 FY 32.7 FZ 0
463 FX 12.5 FY -32.7 FZ 0
464 FX 19.3 FY 83.5 FZ 0.5
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465 FX 19 FY -83.5 FZ -0.1
469 FX 19.6 FY -84.3 FZ 0.1
468 FX 19.2 FY 84.3 FZ -0.6
466 FX 7.8 FY 5.8 FZ -0.1
467 FX 5.1 FY -5.4 FZ 0.2
470 FX 4.9 FY 4.9 FZ 0.1
471 FX 4.5 FY -4.8 FZ -0.1
****
LOAD 4 LOADTYPE Wind  TITLE WIND LOAD (-X DIR)
JOINT LOAD
472 FX -2.1 FY 15.9 FZ 2.2
473 FX -2.2 FY -6.1 FZ -2
475 FX -27 FY 127.4 FZ -1.6
476 FX -2.2 FY 36.4 FZ -2
474 FX -28.9 FY -176.7 FZ 4.9
477 FX -11.6 FY -51.1 FZ 5.9
478 FX -11.6 FY 54.3 FZ -7.4
479 FX -3.2 FY -19.6 FZ 0
480 FX -3.2 FY 19.6 FZ 0
481 FX -2.2 FY -5 FZ 0
482 FX -2.2 FY 5 FZ 0
483 FX -2.2 FY -5 FZ 0
484 FX -2.2 FY 5 FZ 0
485 FX -137 FY -376 FZ 0
486 FX -135 FY 376 FZ 22
487 FX -149 FY -409 FZ 0
488 FX -146 FY 409 FZ -21
460 FX -12.5 FY -34.4 FZ 0
461 FX -12.5 FY 34.4 FZ 0
462 FX -12.5 FY -32.7 FZ 0
464 FX -19.3 FY -83.5 FZ -0.5
465 FX -19 FY 83.5 FZ 0.1
469 FX -19.6 FY 84.3 FZ -0.1
468 FX -19.2 FY -84.3 FZ 0.6
463 FX -12.5 FY 32.7 FZ 0
466 FX -7.8 FY -5.8 FZ 0.1
467 FX -5.1 FY 5.4 FZ -0.2
470 FX -4.9 FY -4.9 FZ -0.1
471 FX -4.5 FY 4.8 FZ 0.1
****
LOAD 5 LOADTYPE Wind  TITLE WIND LOAD(+Z DIR)
****
JOINT LOAD
472 FX 0.3 FY 107.1 FZ 3.6
473 FX 0.1 FY 221.5 FZ 18.4
475 FX 1.4 FY -61.2 FZ 9
476 FX 0.1 FY -222.4 FZ 18.6
474 FX 1.8 FY 25.4 FZ 10.1
477 FX -1.9 FY -28.9 FZ 12.8
478 FX -1.9 FY -41.5 FZ 28.2
479 FX 0.4 FY 3 FZ 0.7
480 FX 0.4 FY -3 FZ 0.8
481 FX -0.1 FY 5.8 FZ 2.2
482 FX -0.1 FY 5.8 FZ 2
```

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483 FX 0.1 FY -5.8 FZ 2.2
484 FX 0.1 FY -5.8 FZ 2
485 FX -16 FY 734 FZ 413
486 FX -18 FY 456 FZ 189
487 FX 14 FY -734 FZ 7
488 FX 19 FY -456 FZ 6
460 FX 0.4 FY 241.7 FZ 20
461 FX -0.4 FY 241.6 FZ 20
462 FX -0.1 FY -241.7 FZ 20.4
463 FX 0.1 FY -241.6 FZ 20.4
464 FX -0.1 FY 196.8 FZ 31.8
465 FX -0.1 FY 192.5 FZ 0.8
469 FX 0.1 FY -192.5 FZ 56.9
468 FX 0.1 FY -196.8 FZ 31.8
LOAD 6 LOADTYPE Wind  TITLE WIND LOAD(-Z DIR)
JOINT LOAD
472 FX -0.3 FY -107.1 FZ -3.6
473 FX -0.1 FY -221.5 FZ -18.4
475 FX -1.4 FY 61.2 FZ -9
476 FX -0.1 FY 222.4 FZ -18.6
474 FX -1.8 FY -25.4 FZ -10.1
477 FX 1.9 FY 28.9 FZ -12.8
478 FX 1.9 FY 41.5 FZ -28.2
479 FX -0.4 FY -3 FZ -0.7
480 FX -0.4 FY 3 FZ -0.8
481 FX 0.1 FY -5.8 FZ -2.2
482 FX 0.1 FY -5.8 FZ -2
483 FX -0.1 FY 5.8 FZ -2.2
484 FX -0.1 FY 5.8 FZ -2
485 FX 15 FY -733 FZ -7
486 FX 20 FY -457 FZ -6
487 FX -16 FY 733 FZ -412
488 FX -18 FY 457 FZ -190
460 FX -0.4 FY -241.7 FZ -20
461 FX 0.4 FY -241.6 FZ -20
462 FX 0.1 FY 241.7 FZ -20.4
463 FX -0.1 FY 241.6 FZ -20.4
464 FX 0.1 FY -196.8 FZ -31.8
465 FX 0.1 FY -192.5 FZ -0.8
469 FX -0.1 FY 192.5 FZ -56.9
468 FX -0.1 FY 196.8 FZ -31.8
LOAD 7 LOADTYPE None  TITLE TEMP LOAD(+25)
JOINT LOAD
466 FX 23.3 FY 20.7 FZ 3.6
467 FX 22.8 FY 15.2 FZ 3.3
470 FX -55.7 FY -20.7 FZ -47.1
471 FX -55.7 FY -7.9 FZ 34.7
LOAD 8 LOADTYPE None  TITLE TEMP LOAD(-25)
JOINT LOAD
466 FX -23.3 FY -20.7 FZ -3.6
467 FX -22.8 FY -15.2 FZ -3.3
470 FX 55.7 FY 20.7 FZ 47.1
471 FX 55.7 FY 7.9 FZ -34.7
****
```

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************************************************************
*****************************************************
**********LOAD COMBS FOR BASE PRESSURE*********************************************
LOAD COMB 201 DL+LL
1 1.0 2 1.0
LOAD COMB 202 DL+0.75*LL+0.75*WL+X
1 1.0 2 0.75 3 0.75
LOAD COMB 203 DL+0.75*LL+0.75*WL-X
1 1.0 2 0.75 4 0.75
LOAD COMB 204 DL+0.75*LL+0.75*WL+Z
1 1.0 2 0.75 5 0.75
LOAD COMB 205 DL+0.75*LL+0.75*WL-Z
1 1.0 2 0.75 6 0.75
LOAD COMB 206 0.6*DL+WL+X
1 0.6 3 1.0
LOAD COMB 207 0.6*DL+WL-X
1 0.6 4 1.0
LOAD COMB 208 DL*0.6+WL+Z
1 0.6 5 1.0
LOAD COMB 209 DL*0.6+WL-Z
1 0.6 6 1.0
LOAD COMB 210 DL+TL+LL
1 1.0 7 1.0 2 1.0
LOAD COMB 211 DL+TL+LL
1 1.0 8 1.0 2 1.0
**********LOAD COMBS FOR MEMB Design*******************************************
LOAD COMB 301 1.4 DL+ 1.7 LL
1 1.4 2 1.7
LOAD COMB 302 1.2 DL+ 0.5 LL+ 1.6 WLX
1 1.2 2 0.5 3 1.6
LOAD COMB 303 1.2 DL+ 0.5 LL+ 1.6 WL-X
1 1.2 2 0.5 4 1.6
LOAD COMB 304 1.2 DL+ 0.5 LL+ 1.6 WL Z
1 1.2 2 0.5 5 1.6
LOAD COMB 305 1.2 DL+ 0.5 LL+ 1.6 WL-Z
1 1.2 2 0.5 6 1.6
LOAD COMB 306 0.9*DL + 1.6 WL+X
1 0.9 3 1.6
LOAD COMB 307 0.9*DL+ 1.6 WL-X
1 0.9 4 1.6
LOAD COMB 308 DL*0.9 + 1.6 WL+Z
1 0.9 5 1.6
LOAD COMB 309 DL*0.9 + 1.6 WL-Z
1 0.9 6 1.6
LOAD COMB 310 1.4 (DL+TL) +1.7 LL
1 1.4 7 1.4 2 1.7
LOAD COMB 311 1.4 (DL+TL) +1.7 LL
1 1.4 8 1.4 2 1.7
************************************************************
PERFORM ANALYSIS PRINT STATICS CHECK
************************************************************
LOAD LIST 301 TO 311
START CONCRETE DESIGN
CODE ACI 2008
```

```
FC 35000 MEMB 1 TO 32 36 TO 61 71 73 TO 83 86 TO 102
FYMAIN 420000 MEMB 1 TO 32 36 TO 61 71 73 TO 83 86 TO 102
TRACK 1 ALL
CLB 0.05 MEMB 71 74 TO 83 86 TO 102
CLT 0.05 MEMB 71 74 TO 83 86 TO 102
CLS 0.05 ALL
MAXMAIN 25 MEMB 1 TO 32 36 TO 61
MINMAIN 16 MEMB 1 TO 32 36 TO 61
MAXMAIN 16 MEMB 71 73 TO 83 86 TO 102
MINMAIN 12 MEMB 71 73 TO 83 86 TO 102
MINSEC 10 ALL
REINF 0 ALL
DESIGN COLUMN 1 TO 32 36 TO 61
DESIGN BEAM 71 73 TO 83 86 TO 102
END CONCRETE DESIGN
FINISH
```

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	18-08-2020	21-10-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	18-08-2020	21-10-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

ATTACHMENT 5 **STAAD OUTPUT OF TOTAL FOUNDATION ARRANGEMENT**

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**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    AUG 18, 2020                *
*           Time=    12:51:30                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\MMGP-2020\ID FAN FDN\SDY-03.08.20\Crusher-Duct-Foundation_File_Final.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 06-AUG-20
    4. JOB PART BAG FILTER _DUCT SUPPORT FOUNDATION
    5. END JOB INFORMATION
    6. INPUT WIDTH 79
    7. UNIT METER KN
    8. JOINT COORDINATES
    9. 1 13.6725 -3.3 -1.2525; 2 14.0725 -3.3 -1.2525; 3 14.5725 -3.3 -1.2525
   10. 4 15.0725 -3.3 -1.2525; 5 15.4725 -3.3 -1.2525; 6 13.6725 -3.3 -0.8525
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230. 632 267 268 253 252; 633 251 252 233 232; 634 252 253 234 233
231. 635 326 327 312 311; 636 327 328 313 312; 637 159 160 144 143
232. 638 160 161 145 144; 639 143 144 130 129; 640 144 145 131 130
233. 641 129 130 116 115; 642 130 131 117 116; 643 115 116 102 101
234. 644 116 117 103 102; 645 101 102 88 87; 646 102 103 89 88; 647 232 233 218 217
235. 648 233 234 219 218; 649 217 218 207 206; 650 218 219 208 207
236. 651 180 181 165 164; 652 164 165 149 148; 653 183 184 168 167
237. 654 167 168 152 151; 655 184 185 169 168; 656 168 169 153 152
238. 657 186 187 171 170; 658 170 171 155 154; 659 187 188 172 171
239. 660 171 172 156 155; 661 190 191 175 174; 662 174 175 159 158
240. 663 178 179 163 162; 664 162 163 147 146; 665 179 180 164 163
241. 666 163 164 148 147; 667 191 192 176 175; 668 175 176 160 159
242. 669 192 193 177 176; 670 176 177 161 160; 671 181 182 166 165
243. 672 165 166 150 149; 673 182 183 167 166; 674 166 167 151 150
244. 675 188 189 173 172; 676 172 173 157 156; 677 189 190 174 173
245. 678 173 174 158 157; 679 210 211 195 194; 680 211 212 196 195
246. 681 225 226 211 210; 682 226 227 212 211; 683 239 240 226 225
247. 684 240 241 227 226; 685 254 255 240 239; 686 255 256 241 240
248. 687 269 270 255 254; 688 270 271 256 255; 689 299 300 285 284
249. 690 300 301 286 285; 691 284 285 270 269; 692 285 286 271 270
250. 693 314 315 300 299; 694 315 316 301 300; 751 357 358 351 350
251. 752 350 351 344 343; 753 343 344 337 336; 754 336 337 330 329
252. 755 362 363 356 355; 756 355 356 349 348; 757 348 349 342 341
253. 758 341 342 335 334; 759 358 359 352 351; 760 351 352 345 344
254. 761 359 360 353 352; 762 352 353 346 345; 763 360 361 354 353
255. 764 353 354 347 346; 765 361 362 355 354; 766 354 355 348 347
256. 767 344 345 338 337; 768 337 338 331 330; 769 345 346 339 338
257. 770 338 339 332 331; 771 346 347 340 339; 772 339 340 333 332
258. 773 347 348 341 340; 774 340 341 334 333; 775 426 427 422 421
259. 776 421 422 417 416; 777 416 417 412 411; 778 411 412 407 406
260. 779 427 428 423 422; 780 422 423 418 417; 781 428 429 424 423
261. 782 423 424 419 418; 783 429 430 425 424; 784 424 425 420 419
262. 787 417 418 413 412; 788 412 413 408 407; 789 418 419 414 413

```
263. 790 413 414 409 408; 791 419 420 415 414; 792 414 415 410 409
264. 795 388 389 383 382; 796 382 383 377 376; 797 389 390 384 383
265. 798 390 391 385 384; 799 391 392 386 385; 800 392 393 387 386
266. 801 383 384 378 377; 802 384 385 379 378; 803 385 386 380 379
267. 804 386 387 381 380; 805 400 401 395 394; 806 394 395 389 388
268. 807 376 377 371 370; 808 370 371 365 364; 809 401 402 396 395
269. 810 395 396 390 389; 811 402 403 397 396; 812 396 397 391 390
270. 813 403 404 398 397; 814 397 398 392 391; 815 404 405 399 398
271. 816 398 399 393 392; 817 377 378 372 371; 818 371 372 366 365
272. 819 378 379 373 372; 820 372 373 367 366; 821 379 380 374 373
273. 822 373 374 368 367; 823 380 381 375 374; 824 374 375 369 368; 837 1 6 7 2
274. 838 6 11 12 7; 839 2 7 8 3; 840 7 12 13 8; 841 11 16 17 12; 842 16 21 22 17
275. 843 12 17 18 13; 844 17 22 23 18; 845 3 8 9 4; 846 8 13 14 9; 847 4 9 10 5
276. 848 9 14 15 10; 849 13 18 19 14; 850 18 23 24 19; 851 14 19 20 15
277. 852 19 24 25 20; 853 26 36 37 27; 854 36 51 52 37; 855 27 37 38 28
278. 856 37 52 53 38; 857 28 38 39 29; 858 38 53 54 39; 859 29 39 40 30
279. 860 39 54 55 40; 861 51 61 62 52; 862 61 71 72 62; 863 52 62 63 53
280. 864 62 72 73 63; 865 53 63 64 54; 866 63 73 74 64; 867 54 64 65 55
281. 868 64 74 75 65; 869 31 41 42 32; 870 41 46 47 42; 871 46 56 57 47
282. 872 56 66 67 57; 873 32 42 43 33; 874 42 47 48 43; 875 47 57 58 48
283. 876 57 67 68 58; 877 33 43 44 34; 878 43 48 49 44; 879 48 58 59 49
284. 880 58 68 69 59; 881 34 44 45 35; 882 44 49 50 45; 883 49 59 60 50
285. 884 59 69 70 60
286. START GROUP DEFINITION
287. ELEMENT
288. _MX 535 TO 550 552 TO 555 565 TO 573 575 576 581 TO 588 590 TO 592 -
289. 594 TO 596 601 602 604 606 608 TO 650 653 TO 660 663 TO 670 679 TO 694 761 -
290. 762 TO 764 769 TO 772 781 782 789 790 797 TO 799 801 TO 803 811 812 819 820 -
291. 837 838 841 842 847 848 851 TO 854 859 TO 862 867 868
292. _MY 521 522 529 530 533 534 537 TO 542 545 TO 558 562 TO 570 575 TO 586 591 -
293. 592 TO 596 602 604 606 608 609 612 615 TO 630 635 636 639 TO 646 649 650 653 -
294. 655 657 659 661 663 665 667 669 675 677 679 TO 682 687 TO 694 751 754 755 -
295. 758 759 761 763 765 768 770 772 774 775 778 779 781 783 788 790 792 -
296. 795 TO 804 837 839 842 844 845 847 850 852 853 855 857 859 862 864 866 868
297. _SHEAR 538 TO 542 545 TO 550 552 TO 555 565 TO 571 573 575 576 578 -
298. 580 TO 584 586 591 592 594 TO 596 601 602 604 615 617 TO 622 624 TO 628 630 -
299. 632 634 TO 636 640 TO 646 648 TO 650 655 657 659 661 663 667 669 675 677 -
300. 679 TO 682 687 TO 694 764 771 781 790 798 802 837 842 847 852 853 859 862 -
301. 868
302. _MX_600 535 TO 543 545 TO 550 552 TO 555 565 TO 571 573 575 576 581 TO 588 -
303. 590 TO 592 594 TO 596 601 602 604 606 608 TO 636 638 640 TO 650 653 TO 659 -
304. 663 TO 670 679 TO 688 691 692
305. _MX_400 761 TO 764 769 TO 772 781 782 789 790 797 TO 799 801 TO 803 811 812 -
306. 819 820 837 838 841 842 847 848 851 TO 854 859 TO 862 867 TO 872 881 TO 884
307. _MY_600 578 580 595 596 598 600 602 604 624 649 650 653 655 657 659 661 663 -
308. 665 667 669 675 677 679 680
309. _MY_400 751 754 755 758 759 761 763 765 768 770 772 774 775 778 779 781 783 -
310. 788 790 792 795 TO 804 837 839 842 844 845 847 850 852 853 855 857 859 862 -
311. 864 866 868 869 872 873 876 877 880 881 884
312. _SHEAR_600 538 TO 542 545 TO 550 552 TO 555 565 TO 570 575 576 578 -
313. 580 TO 584 586 591 592 594 TO 596 602 604 615 617 TO 622 624 TO 628 630 632 -
314. 634 TO 636 640 TO 646 648 TO 650 655 657 659 663 669 679 TO 681 687 688 691 -
315. 692
316. _SHEAR_400 764 771 781 790 798 802 837 TO 839 841 842 844 845 847 848 850 -
317. 851 TO 855 857 859 TO 862 864 866 TO 873 876 877 880 TO 884
318. END GROUP DEFINITION
```

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319. ELEMENT PROPERTY
320. 501 TO 604 606 608 TO 694 THICKNESS 0.6
321. 751 TO 784 787 TO 792 795 TO 824 837 TO 884 THICKNESS 0.4
322. DEFINE MATERIAL START
323. ISOTROPIC CONCRETE
324. E 2.17185E+007
325. POISSON 0.17
326. DENSITY 23.5616
327. ALPHA 1E-005
328. DAMP 0.05
329. TYPE CONCRETE
330. STRENGTH FCU 35000
331. END DEFINE MATERIAL
332. MEMBER PROPERTY AMERICAN
333. 37 TO 44 PRIS YD 1 ZD 0.9
334. 13 TO 16 21 TO 24 PRIS YD 0.6 ZD 0.6
335. 1 TO 8 25 TO 32 PRIS YD 0.7 ZD 0.5
336. 9 TO 12 17 TO 20 PRIS YD 0.55 ZD 0.55
337. 36 45 TO 61 PRIS YD 0.45 ZD 0.45
338. *
339. 71 73 TO 75 78 79 82 83 86 TO 90 PRIS YD 0.6 ZD 0.3
340. 97 TO 102 PRIS YD 0.5 ZD 0.3
341. 76 77 80 81 94 95 PRIS YD 0.4 ZD 0.3
342. 91 TO 93 96 PRIS YD 0.3 ZD 0.3
343. CONSTANTS
344. MATERIAL CONCRETE ALL
345. SUPPORTS
346. 1 TO 430 ELASTIC MAT DIRECT Y SUBGRADE 30000
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**WARNING-CHECK INFLUENCE AREAS FOR ZERO OR LARGE AREAS.

JOINT	INFLUENCE AREA
1	1.889552
2	0.379887
3	0.310789
4	0.179532
5	0.049329
6	0.703316
7	0.202273
8	0.224748
9	0.202274
10	0.089899
11	0.777257
12	0.224748
13	0.249720
14	0.224748
15	0.099888
16	0.703209
17	0.202273
18	0.224748
19	0.202274
20	0.089899
21	1.233807
22	0.359597
23	0.399552

24	0.359598
25	0.159821
26	1.235407
27	0.359597
28	0.399552
29	0.359598
30	0.159821
31	0.128120
32	0.123367
33	0.138042
34	0.129772
35	0.245249
36	0.723785
37	0.202273
38	0.224748
39	0.202274
40	0.089899
41	0.339442
42	0.202273
43	0.224749
44	0.202274
45	0.340945
46	0.577002
47	0.213511
48	0.237235
49	0.213511
50	0.703649
51	0.870147
52	0.224748
53	0.249720
54	0.224749
55	0.099888
56	0.615422
57	0.202273
58	0.224749
59	0.202274
60	0.804658
61	0.808229
62	0.202273
63	0.224748
64	0.202274
65	0.089899
66	0.502133
67	0.260243
68	0.288961
69	0.260206
70	0.596022
71	3.706742
72	1.327996
73	1.002831
74	0.473651
75	0.080086
76	0.168738
77	0.275785
78	0.229821
79	0.266108

80	0.241917
81	0.193533
82	0.774133
83	0.774133
84	0.193533
85	0.241917
86	0.266108
87	0.229821
88	0.275785
89	0.326759
90	0.000000
91	0.542674
92	0.452228
93	0.523632
94	0.476029
95	0.380823
96	1.523294
97	1.523294
98	0.380823
99	0.476030
100	0.523632
101	0.452228
102	0.542673
103	1.405190
104	0.000000
105	0.533777
106	0.444814
107	0.515048
108	0.468226
109	0.374580
110	1.498322
111	1.498322
112	0.374580
113	0.468226
114	0.515048
115	0.444814
116	0.533777
117	1.387452
118	0.003432
119	0.527847
120	0.440885
121	0.514781
122	0.464324
123	0.371459
124	1.485837
125	1.485836
126	0.371459
127	0.464324
128	0.514781
129	0.440885
130	0.527844
131	1.323347
132	0.003404
133	0.511066
134	0.416278
135	0.503281

136	0.476594
137	0.368338
138	1.473351
139	1.473350
140	0.368337
141	0.476594
142	0.503281
143	0.416296
144	0.511073
145	1.307241
146	0.000000
147	0.460729
148	0.365812
149	0.342164
150	0.329485
151	0.323232
152	0.340244
153	1.360976
154	1.360976
155	0.340244
156	0.323232
157	0.329484
158	0.342164
159	0.365813
160	0.460724
161	1.199995
162	0.000000
163	0.421403
164	0.335562
165	0.312150
166	0.280935
167	0.296543
168	0.312150
169	1.248602
170	1.248602
171	0.312150
172	0.296543
173	0.280935
174	0.312150
175	0.335562
176	0.418386
177	1.119813
178	1.456327
179	1.099925
180	0.879034
181	0.817409
182	0.740513
183	0.778607
184	0.817834
185	3.194146
186	3.191624
187	0.662212
188	0.396091
189	0.174321
190	0.181615
191	0.196067

192	0.178441
193	1.765291
194	1.664448
195	0.702766
196	0.793565
197	0.783045
198	0.970904
199	1.204373
200	1.110776
201	3.346542
202	3.363526
203	0.918084
204	0.534282
205	0.408364
206	0.353559
207	0.459646
208	1.922282
209	1.151101
210	0.667488
211	0.252374
212	0.286788
213	0.282964
214	0.376904
215	0.390453
216	0.380674
217	0.294436
218	0.390032
219	1.327463
220	0.482652
221	1.191634
222	0.398410
223	0.483544
224	0.839207
225	0.489807
226	0.177692
227	0.201922
228	0.199230
229	0.273744
230	0.259964
231	0.285734
232	0.207307
233	0.274614
234	1.065633
235	0.803343
236	0.242731
237	0.265635
238	0.271187
239	0.528583
240	0.185443
241	0.210308
242	0.209136
243	0.253029
244	0.306928
245	0.257259
246	0.838592
247	0.894434

248	0.283676
249	0.307212
250	0.260958
251	0.216691
252	0.288415
253	1.260343
254	0.641694
255	0.221221
256	0.253039
257	0.256975
258	0.312284
259	0.358580
260	0.320893
261	0.999469
262	1.011080
263	0.344048
264	0.345158
265	0.310623
266	0.266647
267	0.344139
268	1.708153
269	0.814853
270	0.274571
271	0.316743
272	0.328897
273	0.395317
274	0.419431
275	0.366718
276	1.201935
277	1.221834
278	0.406110
279	0.402465
280	0.376840
281	0.341688
282	0.427870
283	2.141456
284	1.004531
285	0.323076
286	0.372239
287	0.388627
288	0.458861
289	0.496319
290	0.430768
291	1.414042
292	1.437453
293	0.477590
294	0.472908
295	0.440132
296	0.402674
297	0.505684
298	2.260711
299	1.126159
300	0.344614
301	0.397055
302	0.414536
303	0.489452

304	0.529407
305	0.459485
306	1.508311
307	1.533283
308	0.509429
309	0.504436
310	0.469474
311	0.429519
312	0.539396
313	2.103126
314	2.248309
315	0.681736
316	0.785259
317	0.819876
318	0.966444
319	1.052985
320	0.993489
321	4.533143
322	8.259407
323	3.652936
324	3.701960
325	3.523292
326	3.306110
327	3.752029
328	5.937506
329	1.592640
330	0.672777
331	0.671399
332	0.671843
333	0.671764
334	0.671680
335	2.729274
336	0.432213
337	0.187290
338	0.187290
339	0.187290
340	0.187290
341	0.187290
342	1.247900
343	0.444611
344	0.187290
345	0.187290
346	0.187290
347	0.187290
348	0.187290
349	1.336763
350	0.457008
351	0.187290
352	0.187290
353	0.187290
354	0.187290
355	0.187290
356	1.435855
357	3.711234
358	1.386280
359	1.396511

360	1.430840
361	1.489345
362	1.572025
363	14.466751
364	6.782416
365	1.631376
366	1.999055
367	2.103229
368	1.883738
369	12.348152
370	0.833310
371	0.151237
372	0.177926
373	0.177926
374	0.151238
375	0.386123
376	0.911951
377	0.159196
378	0.187290
379	0.187290
380	0.159196
381	0.338265
382	0.928271
383	0.159196
384	0.187290
385	0.187290
386	0.159196
387	0.268338
388	0.673084
389	0.159197
390	0.187290
391	0.187290
392	0.159197
393	0.198413
394	0.343259
395	0.151237
396	0.177926
397	0.177926
398	0.151238
399	0.123723
400	0.061874
401	0.071639
402	0.084281
403	0.084281
404	0.071639
405	0.036579
406	0.384391
407	1.399625
408	1.365296
409	1.355065
410	7.298432
411	0.062430
412	0.124860
413	0.124860
414	0.124860
415	0.629531

416	0.062430
417	0.124860
418	0.124860
419	0.124860
420	0.612244
421	0.062430
422	0.124860
423	0.124860
424	0.124860
425	0.598710
426	0.044928
427	0.172133
428	0.281835
429	0.391538
430	3.072049
TOTAL	322.646851

```
347. MEMBER OFFSET
348. 99 START 0.125 0 0
349. 99 END 0.125 0 0
350. LOAD 1 LOADTYPE DEAD  TITLE DEAD LOAD
**WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
          MASTER/SLAVE OR IF UNCONNECTED JOINTS.
351. JOINT LOAD
352. ****1001 TO 1004 TR-4 OF SECONDARY CRUSHER
353. 472 FX 0 FY -31 FZ -0.2
354. 473 FX 0.1 FY -35 FZ -0.1
355. 475 FX 0 FY -59 FZ -12
356. 476 FX 0.1 FY -35 FZ 0
357. ****1005 TO 1005 TR-3 OF SECONDARY CRUSHER
358. 474 FX -0.1 FY -26 FZ -13
359. 477 FX -0.1 FY -45 FZ 13
360. 478 FX 0.1 FY -46 FZ 14
361. 479 FX 0 FY -26 FZ 0
362. 480 FX 0 FY -26 FZ 0
363. 481 FX 0 FY -6 FZ 0
364. 482 FX 0 FY -6 FZ 0
365. 483 FX 0 FY -6 FZ 0
366. 484 FX 0 FY -6 FZ 0
367. ****2001 TO 2004 BAG FILTER
368. 485 FX -12 FY -635 FZ 0
369. 486 FX 12 FY -653 FZ 0
370. 487 FX -12 FY -636 FZ 0
371. 488 FX 12 FY -654 FZ 0
372. ****3001 TO 3004 TR-5 OF TERTIARY CRUSHER
373. 460 FX -0.1 FY -37 FZ 0
374. 461 FX 0.1 FY -37 FZ 0
375. 462 FX -0.1 FY -37 FZ 0
376. 463 FX 0.1 FY -37 FZ 0
377. ****3011 TO 3014 STAIR CASE
378. 464 FX -0.5 FY -49.4 FZ -3.6
379. 465 FX 0.4 FY -43.8 FZ 0
380. 469 FX 0.5 FY -44.6 FZ 0.1
381. 468 FX -0.4 FY -48.7 FZ 3.6
382. ****5001 TO 5004 3130-FEEDER
383. 466 FX 0.7 FY -21.9 FZ -14.5
384. 467 FX 2.4 FY -47.8 FZ 19.1
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385. 470 FX -1.7 FY -24.5 FZ -2.5
386. 471 FX -3.5 FY -49.2 FZ 5
387. SELFWEIGHT Y -1
388. ELEMENT LOAD
389. 795 TO 824 PR GY -20.2
390. 775 TO 784 787 TO 792 PR GY -15.7
391. 751 TO 774 PR GY -32.5
392. 501 TO 604 606 608 TO 678 681 TO 688 693 694 PR GY -67
393. 837 TO 884 PR GY -67
394. ********************************************************************************
395. LOAD 2 LOADTYPE LIVE  TITLE LIVE LOAD
396. JOINT LOAD
397. 472 FX -0.1 FY 0 FZ 0.3
398. 473 FX 0.2 FY 0 FZ 0
399. 475 FX -0.3 FY 0 FZ 0
400. 476 FX 0.2 FY 0 FZ 0
401. 466 FX 1.2 FY -31.6 FZ -27.5
402. 467 FX 1.2 FY -31.6 FZ 27.5
403. 470 FX -1.9 FY -33.2 FZ -3.9
404. 471 FX -1.9 FY -33.2 FZ 3.9
405. 464 FX -0.1 FY -88.8 FZ -8.5
406. 465 FX -0.1 FY -73 FZ -0.1
407. 469 FX 0.1 FY -74.1 FZ 0.1
408. 468 FX 0.1 FY -87.7 FZ 8.4
409. LOAD 3 LOADTYPE WIND  TITLE WIND LOAD (+X DIR)
410. JOINT LOAD
411. 472 FX 2.1 FY -16 FZ -2.2
412. 473 FX 2.2 FY 6.1 FZ 2
413. 475 FX 27 FY -127.6 FZ 1.6
414. 476 FX 2.2 FY -36.4 FZ 2
415. 474 FX 29 FY 177.1 FZ -4.9
416. 477 FX 11.7 FY 51.2 FZ -5.9
417. 478 FX 11.7 FY -54.4 FZ 7.5
418. 479 FX 3.2 FY 19.7 FZ 0
419. 480 FX 3.3 FY -19.7 FZ 0
420. 481 FX 2.2 FY 5.1 FZ 0
421. 482 FX 2.2 FY -5.1 FZ 0
422. 483 FX 2.2 FY 5.1 FZ 0
423. 484 FX 2.2 FY -5.1 FZ 0
424. 485 FX 138 FY 388 FZ 25
425. 486 FX 140 FY -388 FZ 0
426. 487 FX 143 FY 398 FZ -19
427. 488 FX 145 FY -398 FZ -6
428. 460 FX 12.5 FY 34.4 FZ 0
429. 461 FX 12.5 FY -34.4 FZ 0
430. 462 FX 12.5 FY 32.7 FZ 0
431. 463 FX 12.5 FY -32.7 FZ 0
432. 464 FX 19.3 FY 83.5 FZ 0.5
433. 465 FX 19 FY -83.5 FZ -0.1
434. 469 FX 19.6 FY -84.3 FZ 0.1
435. 468 FX 19.2 FY 84.3 FZ -0.6
436. 466 FX 7.8 FY 5.8 FZ -0.1
437. 467 FX 5.1 FY -5.4 FZ 0.2
438. 470 FX 4.9 FY 4.9 FZ 0.1
439. 471 FX 4.5 FY -4.8 FZ -0.1
440. ****
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441. LOAD 4 LOADTYPE WIND  TITLE WIND LOAD (-X DIR)
442. JOINT LOAD
443. 472 FX -2.1 FY 15.9 FZ 2.2
444. 473 FX -2.2 FY -6.1 FZ -2
445. 475 FX -27 FY 127.4 FZ -1.6
446. 476 FX -2.2 FY 36.4 FZ -2
447. 474 FX -28.9 FY -176.7 FZ 4.9
448. 477 FX -11.6 FY -51.1 FZ 5.9
449. 478 FX -11.6 FY 54.3 FZ -7.4
450. 479 FX -3.2 FY -19.6 FZ 0
451. 480 FX -3.2 FY 19.6 FZ 0
452. 481 FX -2.2 FY -5 FZ 0
453. 482 FX -2.2 FY 5 FZ 0
454. 483 FX -2.2 FY -5 FZ 0
455. 484 FX -2.2 FY 5 FZ 0
456. 485 FX -137 FY -376 FZ 0
457. 486 FX -135 FY 376 FZ 22
458. 487 FX -149 FY -409 FZ 0
459. 488 FX -146 FY 409 FZ -21
460. 460 FX -12.5 FY -34.4 FZ 0
461. 461 FX -12.5 FY 34.4 FZ 0
462. 462 FX -12.5 FY -32.7 FZ 0
463. 464 FX -19.3 FY -83.5 FZ -0.5
464. 465 FX -19 FY 83.5 FZ 0.1
465. 469 FX -19.6 FY 84.3 FZ -0.1
466. 468 FX -19.2 FY -84.3 FZ 0.6
467. 463 FX -12.5 FY 32.7 FZ 0
468. 466 FX -7.8 FY -5.8 FZ 0.1
469. 467 FX -5.1 FY 5.4 FZ -0.2
470. 470 FX -4.9 FY -4.9 FZ -0.1
471. 471 FX -4.5 FY 4.8 FZ 0.1
472. ****
473. LOAD 5 LOADTYPE WIND  TITLE WIND LOAD(+Z DIR)
474. ****
475. JOINT LOAD
476. 472 FX 0.3 FY 107.1 FZ 3.6
477. 473 FX 0.1 FY 221.5 FZ 18.4
478. 475 FX 1.4 FY -61.2 FZ 9
479. 476 FX 0.1 FY -222.4 FZ 18.6
480. 474 FX 1.8 FY 25.4 FZ 10.1
481. 477 FX -1.9 FY -28.9 FZ 12.8
482. 478 FX -1.9 FY -41.5 FZ 28.2
483. 479 FX 0.4 FY 3 FZ 0.7
484. 480 FX 0.4 FY -3 FZ 0.8
485. 481 FX -0.1 FY 5.8 FZ 2.2
486. 482 FX -0.1 FY 5.8 FZ 2
487. 483 FX 0.1 FY -5.8 FZ 2.2
488. 484 FX 0.1 FY -5.8 FZ 2
489. 485 FX -16 FY 734 FZ 413
490. 486 FX -18 FY 456 FZ 189
491. 487 FX 14 FY -734 FZ 7
492. 488 FX 19 FY -456 FZ 6
493. 460 FX 0.4 FY 241.7 FZ 20
494. 461 FX -0.4 FY 241.6 FZ 20
495. 462 FX -0.1 FY -241.7 FZ 20.4
496. 463 FX 0.1 FY -241.6 FZ 20.4
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497. 464 FX -0.1 FY 196.8 FZ 31.8
498. 465 FX -0.1 FY 192.5 FZ 0.8
499. 469 FX 0.1 FY -192.5 FZ 56.9
500. 468 FX 0.1 FY -196.8 FZ 31.8
501. LOAD 6 LOADTYPE WIND  TITLE WIND LOAD(-Z DIR)
502. JOINT LOAD
503. 472 FX -0.3 FY -107.1 FZ -3.6
504. 473 FX -0.1 FY -221.5 FZ -18.4
505. 475 FX -1.4 FY 61.2 FZ -9
506. 476 FX -0.1 FY 222.4 FZ -18.6
507. 474 FX -1.8 FY -25.4 FZ -10.1
508. 477 FX 1.9 FY 28.9 FZ -12.8
509. 478 FX 1.9 FY 41.5 FZ -28.2
510. 479 FX -0.4 FY -3 FZ -0.7
511. 480 FX -0.4 FY 3 FZ -0.8
512. 481 FX 0.1 FY -5.8 FZ -2.2
513. 482 FX 0.1 FY -5.8 FZ -2
514. 483 FX -0.1 FY 5.8 FZ -2.2
515. 484 FX -0.1 FY 5.8 FZ -2
516. 485 FX 15 FY -733 FZ -7
517. 486 FX 20 FY -457 FZ -6
518. 487 FX -16 FY 733 FZ -412
519. 488 FX -18 FY 457 FZ -190
520. 460 FX -0.4 FY -241.7 FZ -20
521. 461 FX 0.4 FY -241.6 FZ -20
522. 462 FX 0.1 FY 241.7 FZ -20.4
523. 463 FX -0.1 FY 241.6 FZ -20.4
524. 464 FX 0.1 FY -196.8 FZ -31.8
525. 465 FX 0.1 FY -192.5 FZ -0.8
526. 469 FX -0.1 FY 192.5 FZ -56.9
527. 468 FX -0.1 FY 196.8 FZ -31.8
528. LOAD 7 LOADTYPE NONE  TITLE TEMP LOAD(+25)
529. JOINT LOAD
530. 466 FX 23.3 FY 20.7 FZ 3.6
531. 467 FX 22.8 FY 15.2 FZ 3.3
532. 470 FX -55.7 FY -20.7 FZ -47.1
533. 471 FX -55.7 FY -7.9 FZ 34.7
534. LOAD 8 LOADTYPE NONE  TITLE TEMP LOAD(-25)
535. JOINT LOAD
536. 466 FX -23.3 FY -20.7 FZ -3.6
537. 467 FX -22.8 FY -15.2 FZ -3.3
538. 470 FX 55.7 FY 20.7 FZ 47.1
539. 471 FX 55.7 FY 7.9 FZ -34.7
540. ****
541. ************************************************************
542. ******************************************************
543. **********LOAD COMBS FOR BASE PRESSURE*****************************************
544. LOAD COMB 201 DL+LL
545. 1 1.0 2 1.0
546. LOAD COMB 202 DL+0.75*LL+0.75*WL+X
547. 1 1.0 2 0.75 3 0.75
548. LOAD COMB 203 DL+0.75*LL+0.75*WL-X
549. 1 1.0 2 0.75 4 0.75
550. LOAD COMB 204 DL+0.75*LL+0.75*WL+Z
551. 1 1.0 2 0.75 5 0.75
552. LOAD COMB 205 DL+0.75*LL+0.75*WL-Z
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553. 1 1.0 2 0.75 6 0.75
554. LOAD COMB 206 0.6*DL+WL+X
555. 1 0.6 3 1.0
556. LOAD COMB 207 0.6*DL+WL-X
557. 1 0.6 4 1.0
558. LOAD COMB 208 DL*0.6+WL+Z
559. 1 0.6 5 1.0
560. LOAD COMB 209 DL*0.6+WL-Z
561. 1 0.6 6 1.0
562. LOAD COMB 210 DL+TL+LL
563. 1 1.0 7 1.0 2 1.0
564. LOAD COMB 211 DL+TL+LL
565. 1 1.0 8 1.0 2 1.0
566. **********LOAD COMBS FOR MEMB DESIGN*****************************************
567. LOAD COMB 301 1.4 DL+ 1.7 LL
568. 1 1.4 2 1.7
569. LOAD COMB 302 1.2 DL+ 0.5 LL+ 1.6 WLX
570. 1 1.2 2 0.5 3 1.6
571. LOAD COMB 303 1.2 DL+ 0.5 LL+ 1.6 WL-X
572. 1 1.2 2 0.5 4 1.6
573. LOAD COMB 304 1.2 DL+ 0.5 LL+ 1.6 WL Z
574. 1 1.2 2 0.5 5 1.6
575. LOAD COMB 305 1.2 DL+ 0.5 LL+ 1.6 WL-Z
576. 1 1.2 2 0.5 6 1.6
577. LOAD COMB 306 0.9*DL + 1.6 WL+X
578. 1 0.9 3 1.6
579. LOAD COMB 307 0.9*DL+ 1.6 WL-X
580. 1 0.9 4 1.6
581. LOAD COMB 308 DL*0.9 + 1.6 WL+Z
582. 1 0.9 5 1.6
583. LOAD COMB 309 DL*0.9 + 1.6 WL-Z
584. 1 0.9 6 1.6
585. LOAD COMB 310 1.4 (DL+TL) +1.7 LL
586. 1 1.4 7 1.4 2 1.7
587. LOAD COMB 311 1.4 (DL+TL) +1.7 LL
588. 1 1.4 8 1.4 2 1.7
589. ******************************************************
590. PERFORM ANALYSIS PRINT STATICS CHECK
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**488**	**NUMBER OF MEMBERS**	**87**
NUMBER OF PLATES	**310**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**426**

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 8, TOTAL DEGREES OF FREEDOM = 1650
TOTAL LOAD COMBINATION CASES = 22 SO FAR.

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 1
LOADTYPE DEAD TITLE DEAD LOAD

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.594810253E+01
Y = -0.181058336E+01
Z = 0.357603277E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = -2.00
SUMMATION FORCE-Y = -11800.28
SUMMATION FORCE-Z = 8.90

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 42201.32 MY= -175.65 MZ= -70188.60

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 2.00
SUMMATION FORCE-Y = 11800.28
SUMMATION FORCE-Z = -8.90

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -42201.32 MY= 175.65 MZ= 70188.60

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 1)

	MAXIMUMS	AT NODE
X	-1.22676E-01	485
Y	-4.82533E-01	90
Z	-2.44853E-01	474
RX	-8.24030E-04	322
RY	-2.40196E-04	474
RZ	-1.01411E-03	1

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 2
LOADTYPE LIVE TITLE LIVE LOAD

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.100050682E+02
Y = 0.349999998E+00
Z = 0.169864742E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = -1.40
SUMMATION FORCE-Y = -453.20
SUMMATION FORCE-Z = 0.20

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 769.90 MY= -192.35 MZ= -4533.81

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 1.40
SUMMATION FORCE-Y = 453.20
SUMMATION FORCE-Z = -0.20

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -769.90 MY= 192.35 MZ= 4533.81

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 2)

	MAXIMUMS	AT NODE
X =	2.27278E-02	464
Y =	-5.26216E-02	5
Z =	-3.75937E-02	466
RX=	-1.83424E-04	466
RY=	2.48979E-05	432
RZ=	-2.44512E-04	1

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
LOADTYPE WIND TITLE WIND LOAD (+X DIR)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.558188912E+01
Y = 0.592591230E+00
Z = 0.364462985E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 816.60
SUMMATION FORCE-Y = 0.50
SUMMATION FORCE-Z = 0.10

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2.73 MY= 2972.96 MZ= -8194.84

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = -816.60
SUMMATION FORCE-Y = -0.50
SUMMATION FORCE-Z = -0.10

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 2.73 MY= -2972.96 MZ= 8194.84

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 3)

	MAXIMUMS	AT NODE
X =	3.49132E-01	485
Y =	1.75846E-01	132
Z =	4.83920E-02	475
RX=	2.57907E-04	314
RY=	1.02253E-04	472
RZ=	-1.01351E-03	485

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 4
LOADTYPE WIND TITLE WIND LOAD (-X DIR)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.554400328E+01
Y = 0.592841407E+00
Z = 0.368315069E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = -817.20
SUMMATION FORCE-Y = -0.40
SUMMATION FORCE-Z = 1.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 2.47 MY= -2972.07 MZ= 8187.10

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 817.20
SUMMATION FORCE-Y = 0.40
SUMMATION FORCE-Z = -1.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -2.47 MY= 2972.07 MZ= -8187.10

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 4)

	MAXIMUMS	AT NODE
X =	-3.42627E-01	485
Y =	-1.71728E-01	132
Z =	-4.98594E-02	472
RX=	-2.64738E-04	314
RY=	-1.01978E-04	444
RZ=	9.95728E-04	485

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 5
LOADTYPE WIND TITLE WIND LOAD(+Z DIR)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.522305412E+01
Y = 0.582025434E+00
Z = 0.169079526E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = -0.30
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 927.70

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 9769.69 MY= -4664.95 MZ= 11.64

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = 0.30
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = -927.70

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -9769.69 MY= 4664.95 MZ= -11.64

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 5)

	MAXIMUMS	AT NODE
X =	-3.72323E-02	464
Y =	1.88683E-01	5
Z =	5.61762E-01	485
RX=	1.90188E-03	485
RY=	4.85234E-04	432
RZ=	7.86951E-04	1

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 6
LOADTYPE WIND TITLE WIND LOAD(-Z DIR)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.523092300E+01
Y = 0.582025434E+00
Z = 0.518276462E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = 0.30
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -927.70

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -9769.69 MY= 4666.75 MZ= -11.64

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = -0.30
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 927.70

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 9769.69 MY= -4666.75 MZ= 11.64

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 6)

	MAXIMUMS	AT NODE
X =	-3.84054E-02	487
Y =	-1.88702E-01	5
Z =	-5.59470E-01	487
RX=	-1.90287E-03	487
RY=	-4.75303E-04	435
RZ=	-7.87086E-04	1

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 7
LOADTYPE NONE TITLE TEMP LOAD(+25)

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = -65.30
SUMMATION FORCE-Y = 7.30
SUMMATION FORCE-Z = -5.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 145.41 MY= -804.36 MZ= 82.42

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = 65.30
SUMMATION FORCE-Y = -7.30
SUMMATION FORCE-Z = 5.50

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -145.41 MY= 804.36 MZ= -82.42

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 7)

	MAXIMUMS	AT NODE
X =	-8.06597E-02	471
Y =	-1.53378E-02	210
Z =	-6.08576E-02	470
RX=	-3.16984E-04	470
RY=	3.54154E-05	470
RZ=	3.15082E-04	471

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 8
LOADTYPE NONE TITLE TEMP LOAD(-25)

```
***TOTAL APPLIED LOAD ( KN   METE ) SUMMARY (LOADING     8 )
    SUMMATION FORCE-X =          65.30
    SUMMATION FORCE-Y =          -7.30
    SUMMATION FORCE-Z =           5.50

   SUMMATION OF MOMENTS AROUND THE ORIGIN-
   MX=        -145.41  MY=         804.36  MZ=         -82.42


***TOTAL REACTION LOAD( KN   METE ) SUMMARY (LOADING     8 )
    SUMMATION FORCE-X =         -65.30
    SUMMATION FORCE-Y =           7.30
    SUMMATION FORCE-Z =          -5.50

   SUMMATION OF MOMENTS AROUND THE ORIGIN-
   MX=         145.41  MY=        -804.36  MZ=          82.42


MAXIMUM DISPLACEMENTS (  CM  /RADIANS) (LOADING      8)
          MAXIMUMS    AT NODE
   X =  8.06597E-02     471
   Y =  1.53378E-02     210
   Z =  6.08576E-02     470
   RX=  3.16984E-04     470
   RY= -3.54154E-05     470
   RZ= -3.15082E-04     471



************ END OF DATA FROM INTERNAL STORAGE ************


591. ******************************************************
592. LOAD LIST 301 TO 311
593. START CONCRETE DESIGN
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594. CODE ACI 2008
595. FC 35000 MEMB 1 TO 32 36 TO 61 71 73 TO 83 86 TO 102
596. FYMAIN 420000 MEMB 1 TO 32 36 TO 61 71 73 TO 83 86 TO 102
597. TRACK 1 ALL
598. CLB 0.05 MEMB 71 74 TO 83 86 TO 102
599. CLT 0.05 MEMB 71 74 TO 83 86 TO 102
600. CLS 0.05 ALL
601. MAXMAIN 25 MEMB 1 TO 32 36 TO 61
602. MINMAIN 16 MEMB 1 TO 32 36 TO 61
603. MAXMAIN 16 MEMB 71 73 TO 83 86 TO 102
604. MINMAIN 12 MEMB 71 73 TO 83 86 TO 102
605. MINSEC 10 ALL
606. REINF 0 ALL
607. DESIGN COLUMN 1 TO 32 36 TO 61
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COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	127.43	44.41	0.11243

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	127.43	2.92	0.00739

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COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	79.69	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	79.69	0.00	0.00000

==========

COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	124.48	5.03	0.01304

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	124.48	2.83	0.00733

==========

COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING